



LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

FAKULTÄT FÜR PHYSIK

MASTERARBEIT IN PHYSIK

**Auf dem Weg zu einem neuen Paradigma eines
physikalischen Wellenfunktionansatzes**

BETREUER

PROF. JUAN CARRASQUILLA-ÁLVAREZ
EIDGENÖSSISCHE TECHNISCHE HOCHSCHULE ZÜRICH

KO-BETREUER

PROF. LODE POLLET
LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

MASTERSTUDENT

TRISTAN ZERWECK

AKADEMISCHES JAHR

2024-2025



LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

DEPARTMENT OF PHYSICS

MASTER THESIS IN PHYSICS

Towards a Novel Paradigm of Physical Wavefunction Approximators

SUPERVISOR

PROF. JUAN CARRASQUILLA-ÁLVAREZ
EIDGENÖSSISCHE TECHNISCHE HOCHSCHULE ZÜRICH

CO-SUPERVISOR

PROF. LODE POLLET
LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

MASTER CANDIDATE

TRISTAN ZERWECK

ACADEMIC YEAR

2024-2025

“I SUPPOSE THEREFORE THAT ALL THINGS I SEE ARE ILLUSIONS; I BELIEVE THAT NOTHING HAS EVER EXISTED OF EVERYTHING MY LYING MEMORY TELLS ME. I THINK I HAVE NO SENSES. I BELIEVE THAT BODY, SHAPE, EXTENSION, MOTION, LOCATION ARE FUNCTIONS. WHAT IS THERE THEN THAT CAN BE TAKEN AS TRUE? PERHAPS ONLY THIS ONE THING, THAT NOTHING AT ALL IS CERTAIN.”

— RENE DESCARTES

THIS TEXT WAS WRITTEN WITHOUT ARTIFICIAL INTELLIGENCE.

Abstract

The quantum many-body problem is exponentially complex with growing system size, requiring an efficient parametrization of the wavefunction to accurately characterize systems with non-trivial correlations. Neural Quantum States (NQS) have emerged as a particularly successful wavefunction ansatz and excel in a wide range of problems, such as modeling ground states, excited states or finite-temperature states and predicting quantum state dynamics. Nevertheless, state-of-the-art NQS architectures, such as vision transformers, are computationally expensive and energy intensive and Variational Monte Carlo (VMC), the typical optimization framework for NQS, is inherently limited by a sampling bottleneck when simulating large system sizes, taking its toll on runtime. Increasing either computational speed, parallelism or reducing energy consumption would benefit the scalability of VMC.

This work proposes a new class of NQS architectures based on analogue computing. Two proof-of-concept devices are evaluated and compared to a standard digital NQS, the restricted Boltzmann machine (RBM). It is shown that for Hamiltonians such as the Heisenberg and Heisenberg J_1 - J_2 models, the analogue NQS outperform the RBM by two orders of magnitude for the same number of parameters in terms of variational ground state energy. Similar results are observed in applications such as predicting the intermolecular equilibrium distance of H_2 and simulating the time-evolution of the transverse field Ising model. These results put photonic computing at the core of an exciting direction to scaling VMC beyond conventional electronic hardware.

Contents

ABSTRACT	vii
LIST OF FIGURES	x
1 INTRODUCTION	i
2 PHOTONIC COMPUTING	5
3 VARIATIONAL MONTE CARLO	9
3.1 Computational Workflow	11
3.2 Markov-Chain Monte Carlo and Metropolis- Hastings Algorithm	14
3.3 Energy Optimization	16
3.4 Hamiltonians	20
4 WAVEFUNCTION ANSÄTZE	23
4.1 Slater-Jastrow Ansatz	23
4.2 Restricted Boltzmann Machine	24
4.3 Analogue Neural Network Ansätze	26
4.3.1 Microring Resonator Meshgrid	28
4.3.2 Optical Reservoir	30
5 RESULTS	35
5.1 Ground State Search	35
5.1.1 Symmetrization Impact	37
5.1.2 Downsampling Effects in Optical Reservoir	38
5.1.3 Influence of Data Replication in Microring Resonator Meshgrid	39
5.2 Time Evolution	41
5.3 H ₂ Molecule	42
5.4 Non-Stabilizerness, Entanglement and Fermionic Antiflatness	43
6 CONCLUSION AND OUTLOOK	45
REFERENCES	47
ACKNOWLEDGMENTS	53
DECLARATION	55

Listing of figures

2.1	Two examples of physical implementations of optical computing	6
3.1	The computational workflow in VMC	13
3.2	Comparison of random sampling and Metropolis-Hastings sampling	16
4.1	Architecture of the RBM	25
4.2	Schematic of a blackbox function. The blackbox can be any physical system	26
4.3	Setup of the Microring Resonator Meshgrid	29
4.4	A chaotic arrangement of LiNbO_3 nanocubes, as seen through scanning electron microscopy	30
4.5	Setup of the Optical Reservoir	31
4.6	Experimental setup of the Optical Reservoir	32
4.7	Four different basis states $ \sigma_i\rangle$ and their corresponding speckle patterns	33
5.1	Relative error of ground state energies for different Hamiltonians on 4×4 lattice	36
5.2	Relative error of ground state energies for different Hamiltonians as well as digital and analogue wavefunctions on a 12×12 and 10×10 lattice.	36
5.3	Improvement of the ground state energy for the J_1 - J_2 model with 4×4 sites with a symmetric wavefunction ansatz	38
5.4	Analysis of average speckle size	38
5.5	Downsampling impact on the relative error of the ground state energy for the TFIM model on a 4×4 lattice.	39
5.6	Exemplary speckle patterns showing different downsampling sizes. The larger the downsampling size, the more pixels are averaged together, resulting in a coarser image and ultimately a loss of information, leading to a lower accuracy of the ground state energy estimation.	39
5.7	Comparison of the optimization behaviour for the Heisenberg model when data replication is used	40
5.8	Comparison of the optimization behaviour for the J_1 - J_2 model when data replication is used	40
5.9	Time evolution of the transverse magnetization for the TFIM model	41
5.10	Infidelity comparison of time evolution for different wavefunctions compared to the exact solution	42
5.11	Energies for different intermolecular distances for the H_2 molecule	42
5.12	Relative errors for different wavefunctions for varying intermolecular distances of the H_2 molecule	43
5.13	Analysis of normalized non-stabilizerness, entanglement entropy and fermionic magic for different wavefunction ansatz initializations	44

1

Introduction

The purpose of science is to create an accurate and predictive understanding of the world. In physics, understanding manifests in deriving theories about nature, including both the world observable by our senses and the hidden, abstract structures found through reasoning [1, 2] or clever experiments [3, 4]. According to Popper a theory is always provisional and only true, as long as it is not falsifiable through empirical evidence [5]. In practice, however, this strong criterion can not be adhered to in physics as many theories are simply unfalsifiable - think of predictions of general relativity about phenomena inside black holes, which are beyond our measurable event horizon and therefore cannot be tested empirically. While scientific progress can be made and better predictions and models are always within reach, theories are only tools with limited scope, never final truths about nature. Similarly, mathematics and physics are powerful, but they might not be fundamental in the sense that they are a complete or final description of reality. The so-called *foundational crisis of mathematics* of the 1920s already exposed the need for a rigorous basis of mathematics grounded on logic and set theory [6] and further fuelled a philosophical debate about the nature of mathematics and its relation to reality. *

This master's thesis is a quest to push the boundaries of a rapidly evolving niche in science, which is computational

*The Vienna Circle of Logical Empiricism should be mentioned in particular for its early work in analytic philosophy, combining formal logic with science [7]. Equally transformative insights were gained in mathematical logic and mathematical philosophy through Gödel's incompleteness theorems, illustrating intrinsic limitations of formal systems, and the development of Zermelo-Fraenkel set theory with the axiom of choice (ZFC), establishing a standard foundation of modern mathematics [8]. Despite many advancements in the past century, several fundamental questions remain open, such as the unprovability of the consistency of ZFC.

quantum physics. The field is in close relation with condensed matter physics, mathematics, computer science, chemistry and material science and this work in specific further overlaps with optics and analogue computing. At the heart of condensed matter physics lies the exploration of microscopic interactions between atoms and electrons which enable the description of various macroscopic phenomena, such as magnetism, (super-) conductivity, quantised Hall conductivity or Bose-Einstein condensation. Insights into those effects have lead to technological breakthroughs, like the developments of semiconductor devices, which are part of everyday digital devices, superconducting magnets for medical imaging, photovoltaic cells for gathering solar energy, batteries, liquid crystal displays (LCDs) and many more [9].

The Schrödinger equation is the cornerstone of solving many of these quantum physical problems. However, since quantum many-body physics bears a computational complexity that scales exponentially with the size of the system, numerical approximations are vital to tackle larger systems with only polynomial complexity. Different computational methods have been developed over the past century to deal with the dimensionality of the Hilbert space, with the earliest one being Hartree in 1928 [10]. The mean-field-theory approach known as the Hartree-Fock method approximates the many-body wavefunction by either a Slater determinant (for fermions) or permanent (for bosons). This description neglects strong correlations between electrons. It is a fast and scalable method often used in chemistry, but some more exotic phases, like spin-liquids, topological states and superconductors can not be accurately modeled (at least without extensions). Density functional theory (DFT) provides the most scalable framework to solve for the electron density and can potentially succeed beyond mean-field-theory if the exchange-correlation functional can be computed accurately, which poses a major challenge in practice [11]. Other methods, like Density matrix renormalization group (DMRG) with tensor networks have had enormous success in the description of strongly correlated 1D gapped systems, however fail to efficiently scale to 2 or 3 dimensions, which are essential to characterize realistic physical systems [12, 13]. DMRG with Matrix-Product-States (MPS), for example, is restricted to area law entangled states in 1D [14] and short-ranged correlations.

The field of machine learning has shown that neural networks are powerful function approximators. In Variational Monte Carlo (VMC) neural network wavefunctions are employed to scale to bigger system sizes in higher dimensions and accurately represent highly entangled ground states and capture large correlation lengths [15]. In addition, these so-called neural quantum states (NQS) have been successfully applied to learn finite temperature states [16], dynamics [15] and excited states [17]. In a machine learning sense this is unsupervised learning of the quantum-many body problem since no training data is needed for VMC. However, NQS are not restricted to unsupervised learning, in fact an early application of NQS to quantum state tomography was to learn an efficient representation of a quantum state from labeled training data [18].

A general trend in the field of machine learning is to use bigger neural networks with more parameters to achieve better performances [19]. This requires more computational resources, faster hardware and more energy. A promising direction of research, which is by no means new [20] but ever more relevant since the success of deep learning, is analogue computing. Photonic computers, for example, are faster [21], can parallel-process efficiently [22] and can be extremely energy efficient using several orders of magnitude less energy than silicon-based computers [23].

The main focus of this work is to apply the ideas of analogue computing to Variational Monte Carlo and parametrize the wavefunction with an analogue neural quantum state. We will outsource a single component of the VMC algorithm to two photonic systems and will characterize the new proof-of-concept physical wavefunctions and compare to a standard NQS, the restricted Boltzmann machine (RBM). The analogue device will transform and extract features that will be used to learn a representation of the wavefunction. We will show that analogue representations are more accurate than the RBM for complex many-body problems with sign structure and also for other applications, like finding the bond equilibrium length of a simple molecule. This is the first analogue wavefunction representation for VMC and opens up an innovative avenue for future large-scale VMC calculations outside the traditional computational hardware paradigm.

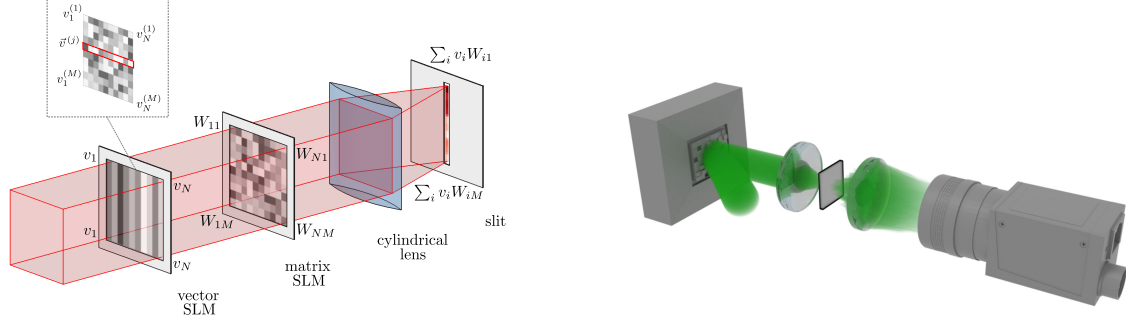
2

Photonic Computing

Since the 1960s, academia has had fluctuating interest in photonic computing [24]. A first strong surge took place in the 1980s when researchers implemented neural networks with optical components for the first time [25, 26]. However, soon the limitations of photonic hardware became apparent, namely that it was only beneficial for large networks and that the implementation of nonlinear activation functions and the control of analogue weights was underdeveloped. Since the era of deep learning in the 2010s a renewed academic and industrial interest emerged due to the increasing demand in compute and energy for modern artificial intelligence applications [22]. At the same time the advancements of complementary metal-oxide-semiconductor (CMOS) technology and computers are reaching a fundamental physical limit, although their development has been arguably the most impactful technological success of the modern era - unchallenged in performance by any other computing paradigm. The size of a single transistor is currently around 1 nm and will soon reach atomic scales at the order of angstroms. This is a dead end for further size reduction and will break Moore's law. Additionally, the sharp increase of required parameters in modern large language models (LLMs) is outpacing the improvement speed of electronic hardware [22, 19].

Hope is growing for the development of special-purpose alternative hardware able to replace digital processors for certain tasks. Significant advancements in photonic components over the past decades driven by the consumer-electronics and optical-communications industries have produced high-resolution cameras at low prices and ultra-high-throughput optical transceivers [22], relevant for smartphones and data centers. This evolution of photonic hardware may finally enable a breakthrough in optical computing. In data centers, for example, optical interconnects are already widespread due to higher bandwidth and lower energy per bit compared to electrical links.

Bringing such developments from applications in communication to computation has the potential of creating processors excelling in terms of speed, energy efficiency and inherent parallelism. Experimental efforts have already realized arbitrary matrix multiplication on easily programmable Mach-Zehnder interferometer meshes [27], on free-space optics setups with spatial light modulators [28] and on microring resonator waveguides [29].



(a) Schematic of a fully reconfigurable coherent optical vector-matrix multiplication. The vector elements are encoded into the laser beam's wavefront amplitude by a spatial light modulator (SLM). The matrix is encoded by a second SLM. A lens at the end sums elements along one dimension, yielding an output vector, which is captured by a camera [28].

(b) Optical Reservoir Computing Using Multiple Light Scattering. The input information is encoded into the light field amplitude or phase through an SLM, like in the neighbouring figure. Upon passing the disordered medium, the photons scatter multiple times and generate random high-dimensional features. The speckle patterns are subsequently recorded by a camera [30].

Figure 2.1: Two examples of physical implementations of optical computing.

Many of these analogue devices, but also some electronic computing systems, share the paradigm of compute-in-memory, where data storage and computation take place in the same physical location [28, 29, 30]. This approach reduces the communication bottleneck deep-rooted in von Neumann architectures, where information has to constantly be channeled between the random access memory (RAM) and the central processing units (CPUs). This is in contrast to graphics processing units (GPUs), who partially address this challenge with integrated video random access memory (VRAM). Photonic computers are naturally compute-in-memory and many operations, such as Fourier transformations, random projections and convolutions can be performed on the fly via the simple propagation of light through free space or light-matter interactions [31]. These transformations can also be reformulated equivalently as matrix-vector multiplications, which are at the heart of a wide range of applications, such as neural networks, scientific computing, combinatorial optimization and cryptography [22]. A stimulating example of optical computing via light-matter interactions is the propagation of light through a dense, disordered medium as seen in Fig. 2.1b, reminiscent of reservoir computing architectures [30]. The scattering process can be modeled by a random matrix multiplication, where the matrix coefficients act as fixed weights that map the input light field into a higher-dimensional feature space. The non-linearity can be realized optically in various ways, such as through the light field intensity measurement by a camera [32] or by using nonlinear media [33].

The expressibility of such optical systems is ultimately dependent on the computational task and the physical

constraints of the analogue device - which is not distinct from digital hardware. Instead of seeing the limitations as disadvantages, one should learn to optimally utilize hardware [34]. A prime example of this are GPUs which are only specialized for certain tasks, including matrix multiplications for image processing or deep learning, where they have become indispensable. But by no means are GPUs general purpose and in fact struggle with, for example, highly sequential logic or irregular data structures.

Two considerations need to be carefully addressed for a realistic treatment of photonic computing. First, the common claim of computing at much greater speeds, namely, the speed of light, is not always justified. While light indeed travels at speed c in vacuum, it propagates at a critically reduced velocity of $0.4c$ in silicon-photonic waveguides. Contrastingly, electronic signals travel at roughly $0.2c$ in CMOS electronic circuits. Therefore, the speed advantage depends on the specific hardware architecture of the optical components. The maximum attainable speedup is around a factor of 5 rather than several orders of magnitude [22].

Second, the energy efficiency of photonic computing is also highly architecture-dependent. While all-optical systems can in principle process information very resource efficiently [23], many photonic accelerator designs require electro-optical conversion, which demands high electrical power consumption. Additionally, absorbing media and non-linear materials can further increase the energy demand. Although there are reduction strategies, one should carefully consider the energy advantages of optical computing in the context of the full system design [31].

Nevertheless, recent experiments demonstrated that when considering larger matrix sizes, larger processor sizes and low-precision arithmetic*, photonic multiply-accumulate operations (MACs) do outperform their electronic counterparts in the metrics of speed, energy efficiency and computational density [35], giving another reason for optimism.

*While analogue computers achieve a maximal precision of around 10 bits, many applications, including neural network inference, are insensitive to low precision [22].

3

Variational Monte Carlo

Quantum mechanical systems in condensed matter physics can be described by the following wave equation. The time-independent and time-dependent Schrödinger equation * are given by

$$\hat{H}\Psi = E\Psi, \quad (3.1)$$

$$\hat{H}\Psi = i\hbar\frac{d}{dt}\Psi, \quad (3.2)$$

with the wavefunction Ψ , the Hamiltonian operator $\hat{H}$, the eigenenergies E and the reduced Planck constant $\hbar$. The solution of these equations yield complete information about the quantum state of a system. While having an accurate description of the physical laws that govern quantum mechanical systems, in fact only small system sizes (two or three dozens of particles) can be exactly diagonalized [39]. Solutions to the Schrödinger equation for large many-body systems are neither obtainable analytically nor solvable via exact diagonalization. Substantial approximations have to be made to find a tractable solution. As Dirac already noted in 1929

*The concept of matter having a wave-like nature - *phase-waves* - was first introduced by deBroglie in 1925 [36]. Erwin Schödinger then postulated the wave equation in 1926 [37]. Dirac later extended the theory to incorporate special relativity [38].

”The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble. It therefore becomes desirable that approximate practical methods of applying quantum mechanics should be developed, which can lead to an explanation of the main features of complex atomic systems without too much computation.” [40]

The complexity of the quantum many-body problem lies in the exponential growth of state space with the system size N , namely of order 2^N . Calculations spanning the whole Hilbert space become untractable. Writing down a state vector Ψ in a computational basis x

$$|\Psi\rangle = \sum_i \Psi_i(x) |x\rangle$$

requires storing 2^N coefficients. For 40 spins the state vector contains around 10^{12} bits which corresponds to 136GByte of memory. Operators acting on such a state would already have 2^{2N} elements, making exact diagonalization infeasible on supercomputing hardware, as the random access memory (RAM) is not sufficiently large. This exponential scaling is reminiscent of the classical n -body problem, which is also not analytically solvable for $n \geq 3$.

Numerical techniques are vital to overcome the curse of dimensionality. Quantum Monte Carlo (QMC) methods leverage smaller parts of the state space to obtain statistically meaningful observables of the system, which avoids exploring the whole Hilbert space.

This work focuses on a specific flavour of QMC, Variational Monte Carlo, which among other things, allows for finding the ground state of a system at zero temperature. The algorithm optimizes a trial wavefunction through stochastically minimizing the energy expectation value, which is lower bounded by the ground state energy. Its success depends on the expressivity of the wavefunction ansatz and the quality of the sampling.

Other quantum Monte Carlo variants, such as Auxiliary Field Monte Carlo [41] and Path Integral Ground State Monte Carlo [42] describe the whole density matrix at finite temperature, but suffer from the notorious sign problem for fermionic systems, making VMC our method of choice.

3.1 COMPUTATIONAL WORKFLOW

There are two key components in Variational Monte Carlo. The first is the variational principle. A variational wavefunction ansatz $\Psi(\theta)$ with parameters θ is chosen and by tuning the parameters in their multi-dimensional parameter space one can minimize the energy and thus find the ground state of the system with parameters θ^* . This corresponds to minimizing a cost function in optimization terms. The variational energy or cost function is expressed through the Rayleigh quotient:

$$E_{\theta^*} = \min_{\theta} \frac{\langle \Psi_{\theta} | H | \Psi_{\theta} \rangle}{\langle \Psi_{\theta} | \Psi_{\theta} \rangle} \quad (3.3)$$

This optimization can also be seen through the lens of Machine Learning. The term "learning" refers to minimizing the cost function iteratively and updating the variational parameters. In practice, the optimization involves numerical techniques adjusted to the non-convex variational energy landscape, which will be discussed in section 3.3.

The second crucial component of Variational Monte Carlo is Monte Carlo Sampling. As described in the previous section, evaluating operators with approximately 2^{2N} elements acting on a state vector with 2^N elements is either computationally expensive or untractable for system sizes of roughly $N > 30$. The role of Monte Carlo sampling is to replace the costly or unfeasible computation of the Rayleigh quotient Eq. (3.3) by a statistical estimate. Instead of summing or integrating over the whole Hilbert-space, expectation values are approximated by averaging over so-called local operators acting on a number of selected samples, which are drawn from a corresponding probability distribution Π_x . For any operator $\hat{O}$ we can write

$$O = \frac{\langle \Psi | \hat{O} | \Psi \rangle}{\langle \Psi | \Psi \rangle}, \quad (3.4)$$

$$= \frac{\sum_{x,x'} \langle \Psi | x \rangle \langle x | \hat{O} | x' \rangle \langle x' | \Psi \rangle}{\sum_x \langle \Psi | x \rangle \langle x | \Psi \rangle}, \quad (3.5)$$

$$= \frac{\sum_{x,x'} |\Psi(x)|^2 O_{xx'} \frac{\Psi(x')}{\Psi(x)}}{\sum_x |\Psi(x)|^2}, \quad (3.6)$$

$$\approx \frac{1}{N_s} \sum_{x \sim \Pi_x} O_{loc}(x), \quad (3.7)$$

$$= \mathbb{E}_{\Pi_x}[O_{loc}(x)], \quad (3.8)$$

where the local operator is defined as

$$O_{loc}(x) = \sum_{x'} O_{xx'} \frac{\Psi(x')}{\Psi(x)}. \quad (3.9)$$

The configuration samples x are now drawn from the probability distribution $\Pi_x \propto |\Psi(x)|^2$. The Monte Carlo sampling is in practice done via Markov Chain Monte Carlo (MCMC). The Markov chain approximation

is unbiased and in the limit of an infinite number of samples it converges to the exact expectation value. Statistical fluctuations decay as $\frac{1}{\sqrt{N_s}}$ allowing for statistical error estimates and the central limit theorem further holds. More light will be shed on the sampling algorithm in the next section (3.2).

The following lines summarize the standard computational steps in VMC.

Algorithm 3.1 Variational Monte Carlo

```

initialize an empty list of samples  $X$ 
while  $\text{length}(X) < N_s$ 
    generate sample  $x_i$  drawn from  $|\Psi_\theta(x)|^2$ 
    if  $\text{acceptance\_rule}(x_i) = \text{true}$ 
        add sample to  $X$ 
    else
        reject sample
    end if
end while
compute  $\langle H \rangle \approx \frac{1}{N_s} \sum_{x \sim \Pi_x} E_{\text{loc}}(x)$  and  $\nabla \langle H \rangle$ 
update parameters of  $\Psi_\theta$  with  $\theta' = \theta - \eta \nabla \langle H \rangle$ 
repeat until convergence

```

The update scheme of the parameters can be as simple and computationally cheap as the Euler method, commonly used for imaginary-time evolution (cf. section 3.3). For real time evolution, however, higher-order integration schemes like Runge-Kutta yield more stable updates [43].

$$\text{Euler method: } \theta' = \theta - \eta \nabla \langle H \rangle \quad (3.10)$$

$$\text{Second order Runge Kutta method (RK2): } \theta' = \theta - \frac{\eta}{2} (\nabla \langle H_\theta \rangle + \nabla \langle H_{\theta+\Delta\theta} \rangle) \quad (3.11)$$

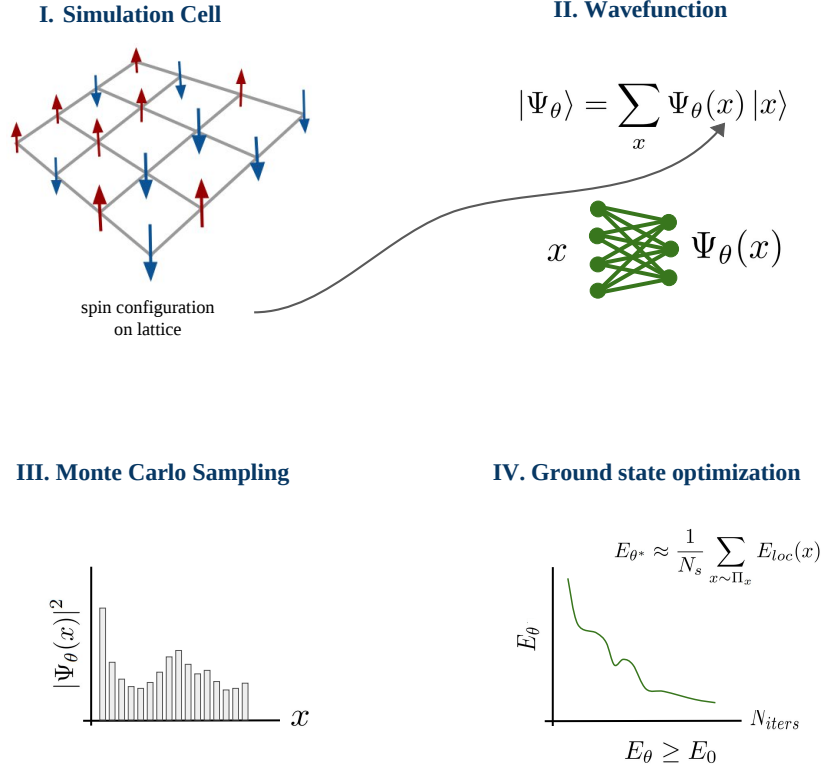


Figure 3.1: The computational workflow in VMC. Subfigure I depicts a randomly generated sample. The computational basis x chosen is the Pauli σ_z basis with spins on a square lattice either pointing up (red) or down (blue). Subfigure II: The chosen sample is then evaluated by the variational wavefunction and the corresponding probability density computed in order to calculate the acceptance probability. If a certain condition is met, explained in the next section, the sample is added to the list of N_s samples. Subfigure III and IV: Once a certain number of samples are generated, which span only a subspace of the Hilbert state space of the quantum many-body system an optimization algorithm, like stochastic gradient descent or stochastic reconfiguration [44] can be executed to approach the ground state iteratively.

3.2 MARKOV-CHAIN MONTE CARLO AND METROPOLIS-HASTINGS ALGORITHM

This chapter derives a Markov-Chain Monte Carlo (MCMC) sampling strategy to obtain representative samples from a high-dimensional probability distribution Π_x that converges to a desired target distribution Π_{eq} . In the context of quantum many-body physics the target distribution we want to approximate is the Born probability density $|\Psi(x)|^2$. In order to achieve this, the MCMC method creates chains of samples $x_1, x_2, x_3, \dots$, which are uniquely characterized by the transition probability $T(x_i \rightarrow x_{i+1})$ of one sample x_i to the next x_{i+1} - also transition probabilities are normalized such that $\sum_i T(x_i \rightarrow x_j) = 1$. The trial distribution Π_x reproduces the target distribution Π_{eq} correctly if it is *stationary*, for which a sufficient condition is the detailed balance condition

$$\Pi_{eq}(x_i)T(x_i \rightarrow x_{i+1}) = \Pi_{eq}(x_{i+1})T(x_{i+1} \rightarrow x_i). \quad (3.12)$$

In other words, a forward process has to be equally as likely as the reverse process if a stationary or equilibrium probability distribution is reached.

To construct a transition probability that satisfies the detailed balance condition and reaches Π_{eq} in a finite number of MCMC steps, one can write the transition probability as the product of the proposal rate P , which is the probability of generating a certain new trial sample, and the acceptance rate A , which determines whether a proposed sample is accepted or rejected.

$$T(x_i \rightarrow x_{i+1}) = P(x_i \rightarrow x_{i+1})A(x_i \rightarrow x_{i+1}). \quad (3.13)$$

By plugging the transition probability into the detailed balance condition (3.12) we find

$$\frac{A(x_i \rightarrow x_{i+1})}{A(x_{i+1} \rightarrow x_i)} = \frac{\Pi_{eq}(x_{i+1})P(x_{i+1} \rightarrow x_i)}{\Pi_{eq}(x_i)P(x_i \rightarrow x_{i+1})}, \quad (3.14)$$

which can be satisfied by the Metropolis-Hastings acceptance rule

$$A(x_i \rightarrow x_{i+1}) = \min \left(1, \frac{\Pi_{eq}(x_{i+1})P(x_{i+1} \rightarrow x_i)}{\Pi_{eq}(x_i)P(x_i \rightarrow x_{i+1})} \right). \quad (3.15)$$

Other acceptance rules can also be derived, as long as they satisfy the detailed balance condition, cf. Glauber or Barker acceptance - however, the Metropolis-Hastings acceptance rule is the most common. The following lines show that the Metropolis-Hastings acceptance rule satisfies the detailed balance condition. In the first case

$$\begin{aligned} \frac{\Pi_{eq}(x_{i+1})P(x_{i+1} \rightarrow x_i)}{\Pi_{eq}(x_i)P(x_i \rightarrow x_{i+1})} &> 1, \\ A(x_i \rightarrow x_{i+1}) &= 1, \\ \Pi_{eq}(x_i)P(x_i \rightarrow x_{i+1})A(x_i \rightarrow x_{i+1}) &= \Pi_{eq}(x_{i+1})P(x_{i+1} \rightarrow x_i)A(x_{i+1} \rightarrow x_i). \end{aligned}$$

Similarly, in the second case

$$\begin{aligned}\frac{\Pi_{eq}(x_{i+1})P(x_{i+1} \rightarrow x_i)}{\Pi_{eq}(x_i)P(x_i \rightarrow x_{i+1})} &< 1, \\ A(x_i \rightarrow x_{i+1}) &= \frac{\Pi_{eq}(x_{i+1})P(x_{i+1} \rightarrow x_i)}{\Pi_{eq}(x_i)P(x_i \rightarrow x_{i+1})}, \\ A(x_{i+1} \rightarrow x_i) &= 1,\end{aligned}$$

where one can again apply the first argument to show that the detailed balance condition is true.

The following algorithm (3.2) summarizes the Metropolis-Hastings algorithm.

Algorithm 3.2 Metropolis-Hastings algorithm

```

generate a sample  $x'$  with the proposal rate  $P(x \rightarrow x')$ 
compute the acceptance rate  $A(x \rightarrow x') = \min\left(1, \frac{\Pi_{eq}(x')P(x' \rightarrow x)}{\Pi_{eq}(x)P(x \rightarrow x')}\right)$ 
pick a uniformly distributed random number  $\alpha \in [0, 1)$ 
if  $\alpha < A(x \rightarrow x')$ 
    add sample  $x'$  to Markov chain as  $x_{i+1}$ 
else
    reject sample and leave  $x_i$  as the last sample in the chain
end if
repeat until  $N_s$  samples in chain

```

In order to understand the phenomenology of the Metropolis-Hastings algorithm it is helpful to compare it to naive random sampling to make its advantages clear. First assume we want to approximate a uniform target distribution with N_s samples. Picking random independent samples uniformly produces an unbiased representation of the underlying probability distribution. However, what if the underlying probability distribution has a structure, for example various local and global maxima, like most physically relevant problems? Would a random choice of samples reflect the characteristics of this probability distribution?

Random sampling will not be efficient and the samples will likely not fall on the maxima in high-dimensional parameter spaces, therefore critical system behaviour will not be captured. Figure 3.2 illustrates the issue. In order to tackle these limitations the Metropolis-Hastings algorithm chooses a biased transition rule to effectively scan more of the high probability states, which dominate the distribution. Configurations x with higher probabilities are favoured against lower weight samples x' due to the acceptance rule (cf. Alg. 3.2). Multiple Markov chains together 'map out' the probability density and are pulled towards different statistically relevant peaks in the state space, because of the intrinsic bias. Despite the advantages of the Metropolis-Hastings algorithm, it is not without limitations. For very high and well separated peaks in the target distributions this and other sampling algorithms can break down. This issue can also arise in ground state search. Some techniques exist to alleviate the problem, like a proper choice of computational basis or importance sampling, which will be addressed later.

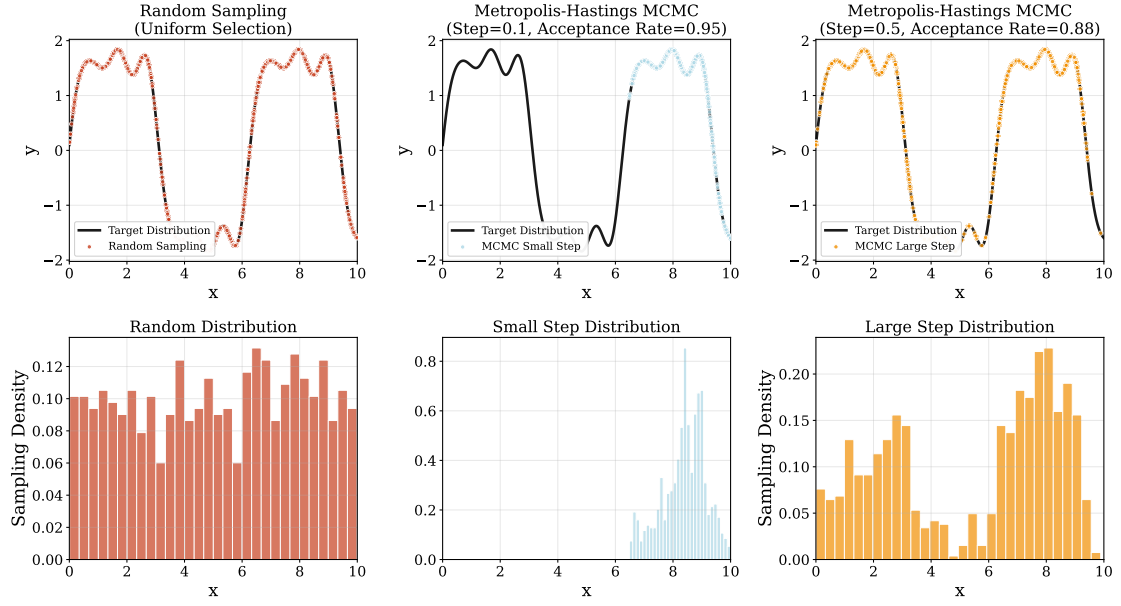


Figure 3.2: Comparison of random sampling and Metropolis-Hastings sampling. In the left subfigures you can see how selecting random points over the target distribution (see lower figure) does not reflect the characteristics of the distribution (upper figure). Instead choosing samples near the maxima captures more relevant behaviour, as seen in the central and right subfigures. The choice of step size in the sampling algorithm is crucial for the quality of the approximation, as it determines the coarsity of the estimated distribution and the range of exploration. A balance has to be struck between exploration and exploitation, i.e. larger and smaller step sizes. Nevertheless, one has to recall that various Markov chains will be deployed to explore the distribution in parallel: for small step size but many chains, all higher probability states will hopefully also be detected (unlike in the central subfigure, where the chain only moves on the right peaks). In the subfigures here only samples from one chain are shown.

3.3 ENERGY OPTIMIZATION

The previous chapter described an iterative way of sampling from the target probability density. Once a sufficient number of samples has been generated, the cost function can be evaluated (Eq. 3.3). Subsequently, the resulting parameter updates can be computed to minimize the variational energy and (imaginary) time-evolve a quantum state. In the following, two important optimization algorithms will be presented, firstly, stochastic gradient descent, the basis of most modern optimization algorithms in machine learning and secondly stochastic reconfiguration, a Riemannian optimization method that accounts for the geometry of the wavefunction manifold.

Let us begin with a concise derivation of the energy gradient. First recall the following expression for the energy

$$\langle \hat{H} \rangle = \frac{\sum_{x,x'} \Psi^*(x) H_{xx'} \Psi(x')}{\sum_x |\Psi(x)|^2} \approx \mathbb{E}_{\Pi}[E_{loc}(x)].$$

Next we can take the derivative with respect to a parameter θ_i

$$\begin{aligned}\partial_{\theta_i} \langle \hat{H} \rangle &= \partial_{\theta_i} \frac{\sum_{x,x'} \Psi^*(x) H_{xx'} \Psi(x')}{\sum_x |\Psi(x)|^2} \\ &= \frac{\sum_{x,x'} \partial_{\theta_i} \Psi^*(x) H_{xx'} \Psi(x') \sum_x |\Psi(x)|^2 + \sum_{x,x'} \Psi^*(x) H_{xx'} \partial_{\theta_i} \Psi(x') \sum_x |\Psi(x)|^2}{(\sum_x |\Psi(x)|^2)^2} \\ &\quad - \frac{\sum_{x,x'} \Psi^*(x) H_{xx'} \Psi(x') \sum_x \partial_{\theta_i} |\Psi(x)|^2}{(\sum_x |\Psi(x)|^2)^2}\end{aligned}$$

Now let us multiply the first term by $\frac{\Psi^*(x)}{\Psi^*(x)}$ and the second by $\frac{\Psi(x')}{\Psi(x')}$:

$$\begin{aligned}\partial_{\theta_i} \langle \hat{H} \rangle &= \frac{\sum_{x,x'} \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \Psi^*(x) H_{xx'} \Psi(x') + \sum_{x,x'} \Psi^*(x) H_{xx'} \Psi(x') \frac{\partial_{\theta_i} \Psi(x')}{\Psi(x')}}{\sum_x |\Psi(x)|^2} \\ &\quad - \frac{\sum_{x,x'} \Psi^*(x) H_{xx'} \Psi(x')}{\sum_x |\Psi(x)|^2} \cdot \frac{\sum_x |\Psi(x)|^2}{\sum_x |\Psi(x)|^2} \left(\frac{\partial_{\theta_i} \Psi(x)}{\Psi(x)} + \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \right)\end{aligned}$$

Next, insert $\frac{\Psi^*(x')}{\Psi^*(x')}$ and $\frac{\Psi(x)}{\Psi(x)}$ into the first and second term, respectively:

$$\begin{aligned}\partial_{\theta_i} \langle \hat{H} \rangle &= \frac{\sum_{x,x'} \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \frac{\Psi^*(x)}{\Psi^*(x')} H_{xx'} |\Psi(x')|^2 + \sum_{x,x'} |\Psi(x)|^2 H_{xx'} \frac{\Psi(x')}{\Psi(x)} \frac{\partial_{\theta_i} \Psi(x')}{\Psi(x')}}{\sum_x |\Psi(x)|^2} \\ &\quad - \frac{\sum_{x,x'} \Psi^*(x) H_{xx'} \Psi(x')}{\sum_x |\Psi(x)|^2} \cdot \frac{\sum_x |\Psi(x)|^2}{\sum_x |\Psi(x)|^2} \left(\frac{\partial_{\theta_i} \Psi(x)}{\Psi(x)} + \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \right) \\ &= \frac{\sum_{x,x'} |\Psi(x')|^2 H_{xx'} \frac{\Psi^*(x)}{\Psi^*(x')} \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)}}{\sum_x |\Psi(x)|^2} + \frac{\sum_{x,x'} |\Psi(x)|^2 H_{xx'} \frac{\Psi(x')}{\Psi(x)} \frac{\partial_{\theta_i} \Psi(x')}{\Psi(x')}}{\sum_x |\Psi(x)|^2} \\ &\quad - \langle \hat{H} \rangle \sum_x |\Psi(x)|^2 \left(\frac{\partial_{\theta_i} \Psi(x)}{\Psi(x)} + \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \right)\end{aligned}$$

$$\begin{aligned}&\approx \mathbb{E}_{\Pi} [E_{loc}(x) \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)}] + \mathbb{E}_{\Pi} [E_{loc}(x) \frac{\partial_{\theta_i} \Psi(x)}{\Psi(x)}] \\ &\quad - \mathbb{E}_{\Pi} [E_{loc}(x)] \mathbb{E}_{\Pi} \left[\frac{\partial_{\theta_i} \Psi(x)}{\Psi(x)} \right] - \mathbb{E}_{\Pi} [E_{loc}(x)] \mathbb{E}_{\Pi} \left[\frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \right] \\ &= \mathbb{E}_{\Pi} \left[E_{loc}(x) \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \right] - \mathbb{E}_{\Pi} [E_{loc}(x)] \mathbb{E}_{\Pi} \left[\frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \right] + h.c.\end{aligned}$$

$$\partial_{\theta_i} \langle \hat{H} \rangle = \mathbb{E}_{\Pi} \left[2 \operatorname{Re} \left[\left(E_{loc}(x) \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} - \mathbb{E}_{\Pi} [E_{loc}(x)] \right) \frac{\partial_{\theta_i} \Psi^*(x)}{\Psi^*(x)} \right] \right] \quad (3.16)$$

Through the expression above, gradients can be efficiently estimated with MCMC. The parameters are then updated via a simple update step, commonly known as stochastic gradient descent.

$$\theta' \leftarrow \theta + \eta \nabla_{\theta} \langle \hat{H} \rangle. \quad (3.17)$$

A more efficient and slightly adapted optimization scheme in variational Monte Carlo is stochastic reconfiguration (SR) [44]. While stochastic gradient descent produces updates in Euclidean parameter space, SR accounts for the Riemannian manifold of the wavefunction and takes the normalization $\int_{-\infty}^{\infty} |\Psi(x)|^2 = 1$ into account. This introduces an extra term, the *quantum geometric tensor* (QGT) or preconditioner, in the parameter update equation 3.18. The QGT ensures the wavefunction parameters are updated in a physically meaningful way on the spherical manifold to preserve the wavefunction's normalization. This is particularly significant for non-linear wavefunctions of strongly-correlated systems, such as those involving Jastrow factors or neural quantum states. In such cases small variational parameter updates can lead to disproportionally large changes in the wavefunction, making stable and metric-aware updates essential.

$$\theta' \leftarrow \theta + \eta Q^{-1} \nabla_{\theta} \langle \hat{H} \rangle. \quad (3.18)$$

In order to understand the exact form of the QGT let us start from the time-dependent Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle.$$

Next we expand the wavefunction in its parameters and in time, such that

$$\begin{aligned} |\Psi(\theta(t + \Delta t))\rangle &\approx |\Psi(\theta(t))\rangle + \Delta t \sum_i \partial_{\theta_i} |\Psi(\theta(t))\rangle \dot{\theta}_i, \\ |\Psi(t + \Delta t)\rangle &\approx |\Psi(t)\rangle - i\Delta t \hat{H} |\Psi(t)\rangle. \end{aligned}$$

We can compute the Pythagorean distance between the two wavefunctions

$$\Delta^2 = \left| \Delta t \sum_i \partial_{\theta_i} |\Psi(\theta(t))\rangle \dot{\theta}_i + i\Delta t \hat{H} |\Psi(t)\rangle \right|^2.$$

If this distance vanishes then our variational wavefunction is a solution of the time-dependent Schrödinger equation. Therefore, we can minimize this distance by taking the derivative with respect to the parameters θ . When taking proper normalization into account this leads to a set of equations for the parameter updates $\dot{\theta}_k$.

$$\sum_j \text{Re}(Q_{i,j}) \dot{\theta}_j(t) = \text{Im}(F_i), \quad (3.19)$$

with

$$Q_{i,j} = \frac{\langle \partial_{\theta_i} \Psi | \partial_{\theta_j} \Psi \rangle}{\langle \Psi | \Psi \rangle} - \frac{\langle \partial_{\theta_i} \Psi | \Psi \rangle \langle \Psi | \partial_{\theta_j} \Psi \rangle}{\langle \Psi | \Psi \rangle^2},$$

$$F_i = \frac{\langle \partial_{\theta_i} \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} - \frac{\langle \partial_{\theta_i} \Psi | \Psi \rangle \langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle^2}.$$

The parameter updates can be solved iteratively with the Euler method or the more stable Runge-Kutta method.

$$\theta_k(t + \Delta t) = \theta_k(t) + \Delta t \dot{\theta}_k(t).$$

While the solution of the time-dependent Schrödinger equation is a more general framework to do dynamics of quantum states, we will also use the time-independent Schrödinger equation to compute ground states of quantum many-body systems. Let us derive the equations for the parameter updates of the so-(misleadingly)-called imaginary time evolution. The first step is to rewrite $\tau = it$ in the Schrödinger equation

$$|\Psi(\tau)\rangle = e^{-\tau \hat{H}} |\Psi(0)\rangle. \quad (3.20)$$

This evolution is no longer unitary and will converge to the ground state of the Hamiltonian, which can be seen from the spectral decomposition

$$|\Psi(\tau)\rangle = \sum_k e^{-\tau E_k} |\Psi_k\rangle,$$

$$|\Psi(\tau)\rangle = e^{-\tau E_0} \left(|\Psi_0\rangle + \sum_{k>0} e^{-\tau(E_k - E_0)} |\Psi_k\rangle \right),$$

where $|\Psi_k\rangle$ are the eigenstates of the Hamiltonian and E_k are the corresponding eigenvalues. When factoring out the ground state energy term, we find that the higher excited states will be exponentially suppressed for large τ and we will converge to the exact ground state wavefunction. From this one can deduce that VMC only converges for systems with a gap between the ground state and higher excited states.

Following the same logic as for the time-dependent Schrödinger equation, we can derive the parameter updates for the imaginary time evolution. First expanding the wavefunction in its parameters and time, we find

$$|\Psi(\theta(\tau + \Delta\tau))\rangle \approx |\Psi(\theta(\tau))\rangle + \Delta\tau \sum_i \partial_{\theta_i} |\Psi(\theta(\tau))\rangle \dot{\theta}_i,$$

$$|\Psi(\tau + \Delta\tau)\rangle \approx |\Psi(\tau)\rangle - \Delta\tau \hat{H} |\Psi(\tau)\rangle.$$

After taking the derivative of the distance we get slightly modified equations for the parameter updates.

$$\sum_j \text{Re}(Q_{i,j}) \dot{\theta}_j(\tau) = \text{Re}(F_i),$$

with the same quantum geometric tensor and force vector as above.

3.4 HAMILTONIANS

At the heart of condensed matter physics lies the modeling of materials on a microscopic level to understand emergent macroscopic properties. Fundamental questions are: Which interactions govern a system's behaviour? Are these interactions short-ranged or long-ranged? Do certain ordering patterns emerge at larger scales? Where do phase transitions occur?

To address these questions, theorists have come up with effective lattice models, that grasp essential physics without regarding all microscopic degrees of freedom. Notable contributions were made by Lenz and Ising [45], Heisenberg [46] and Hubbard, Gutzwiller and Kanamori [47] to name a few. Their models successfully describe various quantum phases, such as quantum magnetism, superconductivity and charge-density waves as well as phase transitions at low and high temperatures, specifically in regimes where statistical mechanics fails. The main assumptions are that fermionic or bosonic particles are localized at lattice sites and interact with their nearest or next-nearest neighbours only. While some exact analytical solutions exist to these models in one dimension, most solutions remain intractable for larger system sizes [46], which is why numerical approximations are necessary.

Since these fundamental models have been of great interest to the condensed matter community, they are excellent benchmarks for our VMC calculations with our new physical NQS approach. We consider three representative Hamiltonians and use the notation $\sigma^x, \sigma^y, \sigma^z$ to represent the spin degrees of freedom (Pauli matrices) and set $\hbar = 1$.

Transverse Field Ising model

$$\hat{H} = -J \sum_{\langle i,j \rangle} \hat{\sigma}_i^z \hat{\sigma}_j^z - h \sum_i \hat{\sigma}_i^x. \quad (3.21)$$

Heisenberg model

$$\hat{H} = \frac{J}{2} \sum_{\langle i,j \rangle} \hat{\boldsymbol{\sigma}}_i \cdot \hat{\boldsymbol{\sigma}}_j, \quad (3.22)$$

with

$$\hat{\boldsymbol{\sigma}}_i = \frac{1}{2} (\hat{\sigma}_i^x, \hat{\sigma}_i^y, \hat{\sigma}_i^z)^T.$$

This can be conveniently rewritten in terms of $\sigma^\pm = \frac{1}{2}(\sigma^x \pm i\sigma^y)$ to describe an effective hopping model, reminiscent of the Hubbard model.

$$\hat{H} = J \sum_{\langle i,j \rangle} (\hat{\sigma}_i^+ \hat{\sigma}_j^- + \hat{\sigma}_i^- \hat{\sigma}_j^+ + \hat{\sigma}_i^z \hat{\sigma}_j^z) \quad (3.23)$$

$J_1 - J_2$ **Heisenberg model**

$$\hat{H} = \frac{J_1}{2} \sum_{\langle i,j \rangle} (\hat{\sigma}_i \cdot \hat{\sigma}_j) + \frac{J_2}{2} \sum_{\langle\langle i,j \rangle\rangle} \hat{\sigma}_i \cdot \hat{\sigma}_j, \quad (3.24)$$

which includes second-nearest neighbour interactions.

4

Wavefunction Ansätze

The choice of an appropriate variational wavefunction ansatz is crucial for the success of Variational Monte Carlo (VMC). A flexible ansatz is necessary to accurately describe the true ground state, allowing for the correct estimation of observables and characterization of quantum phases. In general, the expressivity of an ansatz is directly related to the accuracy of the calculation, if there is no prior knowledge of the ground state wavefunction. An ansatz capable of representing multiple quantum phases is desirable as it generalizes well across different systems and Hamiltonians. Integrating physically motivated constraints or symmetries, such as conservation laws, spin symmetries or translational invariance, can significantly boost VMC performance, as it keeps the optimization on a reduced and physically meaningful variational manifold.

This chapter first introduces the Slater-Jastrow ansatz, a well-established and computationally efficient ansatz for quantum many-body physics. Subsequently, neural quantum states (NQS) are discussed with a focus on the Restricted Boltzmann Machine (RBM). Finally, two novel analogue wavefunction ansätze inspired by photonic computing are proposed and analyzed: the Microring Resonator Meshgrid (MRM) and the optical reservoir (OR).

4.1 SLATER-JASTROW ANSATZ

The Slater-Jastrow ansatz consists of a fermionic antisymmetric determinant of single-particle orbitals and a Jastrow factor accounting for many-body interactions beyond a mean-field description of the system. It is written as

a product of a Slater determinant and a correlation factor.

$$|\Psi\rangle = |\Psi_{\text{Slater}}\rangle |\Psi_{\text{Jastrow}}\rangle. \quad (4.1)$$

The Slater determinant can be expressed as

$$\Psi(\{\mathbf{r}\}) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} \phi_1(\mathbf{r}_1) & \phi_2(\mathbf{r}_1) & \phi_3(\mathbf{r}_1) & \cdots & \phi_N(\mathbf{r}_1) \\ \phi_1(\mathbf{r}_2) & \phi_2(\mathbf{r}_2) & \phi_3(\mathbf{r}_2) & \cdots & \phi_N(\mathbf{r}_2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{r}_N) & \phi_2(\mathbf{r}_N) & \phi_3(\mathbf{r}_N) & \cdots & \phi_N(\mathbf{r}_N) \end{pmatrix}, \quad (4.2)$$

where the determinant embodies the particle-exchange antisymmetry (Pauli principle). The terms $\phi_i(\mathbf{r}_i)$ inside the determinant are the single-particle orbitals. For bosonic systems the determinant is replaced by a permanent. Unlike the determinant, which is efficiently computable in polynomial time, the permanent is exponentially-hard to compute, constraining the size of simulatable bosonic systems [46].

The Jastrow factor contains specific many-body correlations and can be expressed in the following form

$$J(\{\mathbf{r}\}) = \exp \left(\sum_i u_1(\mathbf{r}_i) + \sum_{i<j} u_2(\mathbf{r}_i, \mathbf{r}_j) + \sum_{i<j<k} u_3(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \dots \right), \quad (4.3)$$

with u_i the single and many-body correlation functions. In practice, the interaction functions are approximated by polynomials (e.g. splines) or neural networks, which variationally learn the optimal correlations. Usually, the Jastrow factor is truncated at two-body interactions, because of the computational overload for higher order interactions.

This parameter efficient ansatz has proved to be highly successful for various problems in quantum many-body physics and quantum chemistry and the correlations can be easily interpreted. In more recent years, however, novel neural network based wavefunction ansätze achieved lower ground state energies than traditional models [48] and seem to have higher representational power. On the downside, they lack interpretability, especially the more complex the architecture becomes.

4.2 RESTRICTED BOLTZMANN MACHINE

The earliest neural network architecture used to parametrize quantum many-body states is the Restricted Boltzmann Machine (RBM) [15]. Its structure is simple compared to modern day neural network architectures basing on the transformer [49]. Nevertheless, the RBM was successfully applied to e.g. the Ising and Heisenberg model in one and two spatial dimensions to explore ground states and dynamics of the respective systems. The RBM

wavefunction can be expressed as a sum over the hidden configurations h

$$\Psi(v) = \sum_h e^{E(v,h)}, \quad (4.4)$$

$$E(v, h) = \sum_i a_i v_i + \sum_j b_j h_j + \sum_{ij} W_{ij} v_i h_j, \quad (4.5)$$

where the input v is a binary lattice spin configuration and h is a hidden configuration. Further, the factors a_i are the so-called visible biases and b_j the hidden biases, while W_{ij} are the weights, which connect the visible and hidden layers. Since the hidden variables introduce non-local correlations between the input particles, the RBM can represent long-range correlated quantum phases.

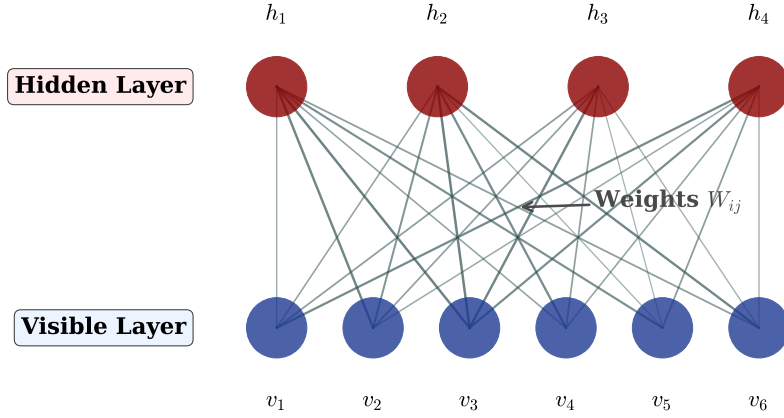


Figure 4.1: Architecture of the RBM. There are two fully connected layers, the visible and hidden layer. The output is the exponential of the energy of the overall neuron configuration, which is the sum over all neurons, their biases and their weighted connections.

It was demonstrated by Carleo and Troyer [15] that for an increased dimension of the hidden layer, the ground state accuracy increases, which is expected for a neural network's expressive power according to the universal approximation theorem [50]. Additionally, the RBM ansatz has shown to outperform tensor network methods, such as Matrix Product State (MPS) [15], underlining the potential of neural quantum states to compactly represent highly entangled quantum many-body states in a polynomial number of parameters.

4.3 ANALOGUE NEURAL NETWORK ANSÄTZE

The following two subsections present a novel class of neural quantum states, that we term analogue NQS. These architectures have the objective to exceed digital NQS in either terms of accuracy, computational speed or energy efficiency by profiting from physical dynamics in analogue devices.

To understand what constitutes a suitable analogue device to be used as a quantum state parametrization in Variational Monte Carlo, it is useful to examine the functional nature of a wavefunction ansatz. Any function is a mapping from an input space $\mathcal{X}$, the domain, to an output space $\mathcal{Y}$, the codomain,

$$f : \mathcal{X} \rightarrow \mathcal{Y}, x \mapsto f(x)$$

where every element in the domain maps to a unique element in the codomain. The evolution of many physical systems is described by partial differential equations, whose solutions are such functions that map an initial state to a future state. We can design analogue devices that apply such controlled transformations to some input information directly in hardware. The physical transformation is the computation and therefore every physical system can be seen as a computer. This is at the heart of analogue computing and goes beyond von Neumann architectures and the usual notion of computation, which bases on bits and logic gates [51].

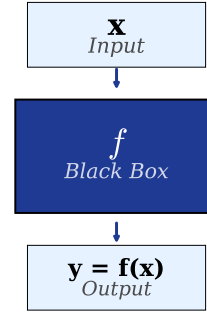


Figure 4.2: Schematic of a blackbox function. The blackbox can be any physical system

Further, the physical system does not have to be quantum mechanical in nature to represent the target quantum state wavefunctions, such as for example digital neural quantum states also materialize on classical digital bits. And notably, the whole wavefunction does not have to simultaneously exist on an analogue device for VMC, but it is sufficient to produce instances of the wavefunction $\Psi(x)$ which are the coefficients corresponding to single basis states $|x\rangle$. Recall, the total wavefunction can be formulated as

$$|\Psi\rangle = \sum_x \Psi(x) |x\rangle,$$

and each of these coefficients could be realized by a "black-box" analogue device as desired outputs $\Psi(x)$ of the input basis states x_i .

As an illustrative example, consider a square metal plate. Imagine propagating some information through this system. The input could for example be encoded in the frequencies of an electronic signal, an audio signal, or a mechanical operation on the plate. The information will be deterministically processed by the analogue device and create a local and unique response - our computation. The response can be measured through physical quantities, such as vibrational modes or electrical signals of the plate. In the context of VMC this output signal needs to be decoded and can then be used to represent a complex value for the wavefunction coefficient. Through Markov

Chain Monte Carlo sampling the wavefunction can further be evaluated at a sufficient number of samples and the gradients of the wavefunction can be computed to iteratively update the parameters of the wavefunction in order to improve the representation of the quantum state.

A central aspect for this procedure are tunable parameters in the analogue device and knowledge of the gradients of the system with respect to those parameters. In practice, adding tunable parameters is easily implementable since one can usually expand the input space and define some controllable variables as parameters. Returning to the example of the square metal plate, one could for example define a certain range of input electronic frequencies to encode the input information and another range to encode the parameters. Nevertheless, finding gradients of analogue devices is challenging, since the exact transformations inside physical systems are often unknown.

We adopt two distinct approaches to solve this issue. The first approach is to work with an analogue device that is fully understood mathematically. For the Microring Resonator Meshgrid the partial differential equations are known that govern the system and allow to analytically differentiate the Rayleigh quotient (Eq. 3.3) with respect to system parameters.

The second solution is a hybrid approach, namely adding a digital layer on top of the analogue device. By feeding the output of a black box device into a single feedforward layer and differentiating with respect to the digital weights, one can learn to extract the right features from the blackbox. This delegates the optimization task to the digital component, while keeping the higher order feature extraction inside the analogue device.

4.3.1 MICRORING RESONATOR MESHGRID

In 2024 Wanjura et al. [29] proposed a scheme to do nonlinear neuromorphic computing with a system composed of stacked racetrack resonators. The microring resonators are closed-loop waveguides with a diameter of a few micrometers making them suitable for integrating them densely on photonic chips. The resonators are coupled to a source, allowing for light to circulate within the ring at wavelengths of integer multiples of the ring's circumference. The dynamics of the system can be described by linear coupled mode equations in the frequency domain

$$-i\omega a_j(\omega) = -i \sum_l H_{j,l}(x, \theta) a_l(\omega) - \sqrt{\kappa_j} a_{j,probe}(\omega). \quad (4.6)$$

where $\mathbf{a}(\omega)$ is the vector of N resonator mode amplitudes, $H(x, \theta)$ is the Hamiltonian of the system and κ is the coupling strength to an external probe signal. This equation can in turn be rewritten in terms of a scattering matrix $S(\omega, x, \theta)$, such that

$$\mathbf{a}_{response}(\omega) = S(\omega, x, \theta) \mathbf{a}_{probe}(\omega), \quad (4.7)$$

with the scattering matrix given by

$$S(\omega, x, \theta) = \mathbf{1} - i\kappa(\omega - H(x, \theta))^{-1}, \quad (4.8)$$

$$= \mathbf{1} - i\kappa G(\omega, x, \theta), \quad (4.9)$$

where x is our input data. The Hamiltonian includes all relevant information about coupling strengths, decay rates and detunings of the resonators. The Green's function $G(\omega, x, \theta)$ is the inverse of the Hamiltonian shifted by the resonator frequency ω , and can further be expressed as

$$G(\omega, x, \theta) = \left(\frac{\kappa_{tot}}{2} + i(\Delta + x - \omega) \right)^{-1}. \quad (4.10)$$

Interestingly, this is reminiscent of a one-layer feedforward neural network, which applies an activation function to a linear transformation of an input. Formally, one expresses this as $f(x) = \sigma(Wx + b)$, where W is the weight matrix, b is the bias vector and σ is the activation function. In the MRM we have bias terms inside the inversion function, $\frac{\kappa_{tot}}{2}$, $i(\Delta - \omega)$, and the non-linear activation and weight matrix are expressed through the inverse function. Although the relation between probe signal and response is linear, as seen from Eq. 4.7, the parameters of the system are transformed non-linearly, as seen from the scattering matrix (Eq. 4.8), and the Green's function (Eq. 4.10). The Hamiltonian matrix is inverted to obtain the Green's function, which is a non-linear operation. Therefore, if we encode our input in the parameters of this equation, namely the detunings, one can efficiently generate higher order features with linear wave scattering. The concept of using linear systems for non-linear transformations is also present in other applications, such as variational quantum circuits [52].

Concludingly, one can measure the response modes for a given probe signal and then use the ratio of response

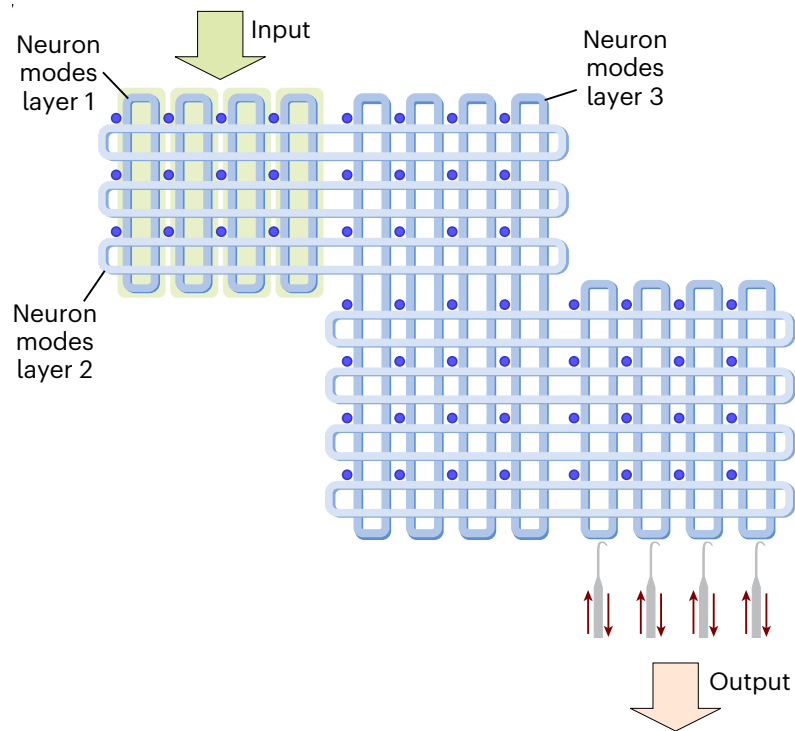


Figure 4.3: Setup of the Microring Resonator Meshgrid. Every resonator is driven by a laser and coupled to its neighbouring resonators. The coupling strength depends on the optical intensities in the neighbouring resonators. Light scatters linearly through the grid and can be measured via output detectors. Taken from [29].

to probe signal to compute the scattering matrix elements (Eq. 4.7). Additionally, the gradients with respect to tunable physical parameters can also be derived analytically, cf. [29] and can even be experimentally read out by measuring the scattering matrix elements and combining them as follows:

$$\frac{\partial S_{r,p}}{\partial \Delta_j} = i\sqrt{\kappa_p \kappa_r} G_{j,p} G_{r,j}, \quad (4.11)$$

$$\frac{\partial S_{r,p}}{\partial J_{j,l}} = i\sqrt{\kappa_p \kappa_r} (G_{j,p} G_{r,l} + G_{l,p} G_{r,j}). \quad (4.12)$$

The derivative with respect to the detuning on site j depends on the scattering from the probe site p to site j and from j to the response site r and similarly for the derivative with respect to the coupling strength. When measuring all response mode amplitudes, which requires N_r measurements, one can characterize the whole scattering matrix and obtain all gradients.

Thus, the microring resonator meshgrid represents a physical realizable system capable of parametrizing a variational wavefunction. It enables encoding some input, transforming non-linearly through the scattering dynamics and measuring outputs and gradients in parallel, thereby facilitating efficient variational parameter optimization and wavefunction learning.

4.3.2 OPTICAL RESERVOIR

The second physical device we propose as an analogue wavefunction ansatz is the Optical Reservoir (OR). In contrast to the Microring Resonator Meshgrid the internal dynamics are completely unknown. The system consists of a disordered arrangement of LiNbO_3 nanocubes with sizes between 100 and 400 nm, which scatter light multiple times to generate complex non-linear output speckle patterns with high-dimensional features. Due to the chaotic nature of the system, its transformation is intractable and therefore the entire analogue device is treated as a black box.

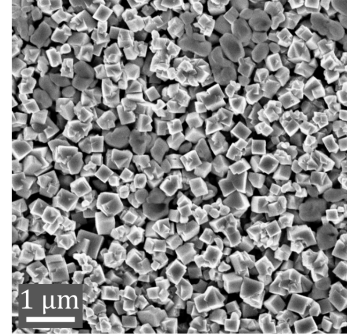


Figure 4.4: A chaotic arrangement of LiNbO_3 nanocubes, as seen through scanning electron microscopy. Taken from [53].

When light passes through the disordered medium, it undergoes second harmonic generation (SHG) - a non-linear optical effect, where two photons combine to form a photon with double the frequency, allowing for the generation of quadratic features. The generation of higher order features is in turn beneficial for the expressivity of the wavefunction ansatz.

Due to the abundance of disorder, no exact analytical description exists of either the optical reservoir or the gradients with respect to the system parameters. While, in principle, numerical techniques could be employed to approximate the system's response and compute gradient estimates [54], we will instead adopt a pragmatic approach and process the reservoir output speckles by an additional digital linear layer. This enables the computation of gradients with respect to digital weights. The learning of a quantum state in VMC then directly translates to the learning of these digital weights.

Although disordered media can often be described by diffusive theory, recent findings suggest that the general

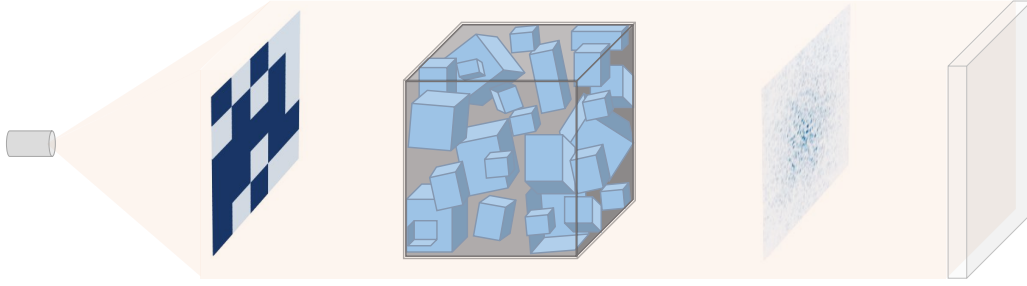


Figure 4.5: Setup of the Optical Reservoir. The input light field hits an SLM which encodes the basis state and propagates through the disordered medium non-linearly. The output speckle pattern is subsequently digitized by a camera and processed by a linear neuron layer to produce a scalar output value.

dynamics of this system can be described by mesoscopic transport [53], which might open a pathway to deriving analytical expressions for the gradients in the future.

The input to the OR is encoded in the phase of the incoming light field via a spatial light modulator (SLM). Other encoding schemes are in principle also feasible, for example encoding data in the light field amplitudes, and should be explored in future work. The light then passes through the non-linear disordered medium (NLDM) and is captured by a camera. Finally, the digital speckle pattern is processed by a linear neuron layer. The conceptual design is depicted in Fig. 4.5, while the experimental realization is shown in Fig. 4.6. Each input basis state generates a deterministic and unique speckle pattern, as illustrated in Fig. 4.7. These patterns serve as high-dimensional representations of input basis states and together the whole set defines the quantum state. It should be noted that the camera introduces an additional non-linearity, since it does an intensity measurement of the transmitted speckle patterns, which captures the squared modulus of the light field amplitude. The exact effects of the camera and its implications, however, remain an object of future investigation. This architecture, analogously to the MRM, is a physical implementation of a neural quantum state, capable of computing wavefunction coefficients $\Psi_i(\sigma_i)$ for individual basis states $|\sigma_i\rangle$ in the context of VMC. The inclusion of a digital linear layer elegantly circumvents the challenge of hidden internal dynamics of the reservoir and unknown gradients. Simultaneously, the higher order feature extraction and implicit 'machine learning' reside inside the black box. The linear layers merely select the relevant features from the speckle patterns for the optimization task, preserving the analogue device's role as a feature extractor.

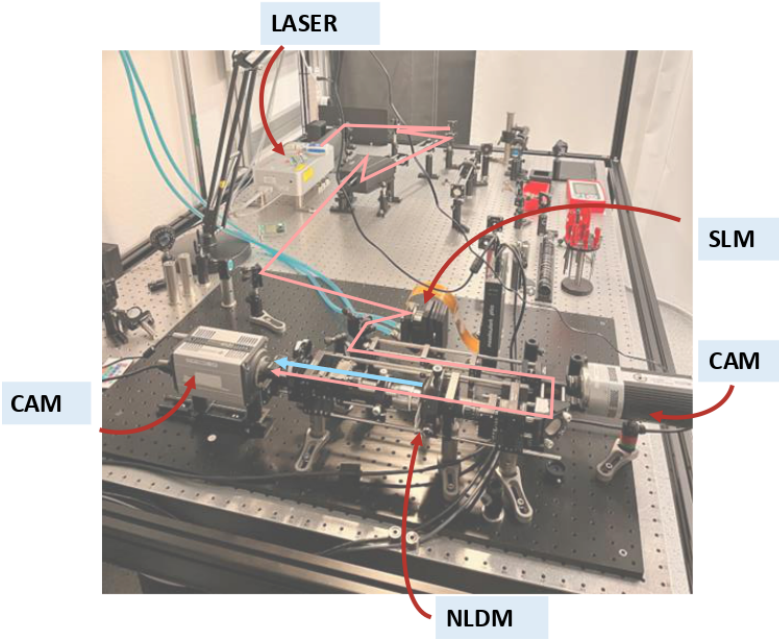


Figure 4.6: Experimental setup of the Optical Reservoir. The light beam is shown in a rose color and starts from the laser (LASER) on the top left. The collimated light beam hits the spatial light modulator (SLM), which imprints some information, in our case the basis state $|\sigma_i\rangle$, onto the light field. A beam splitter leads one beam to a reference camera (CAM) to measure the accuracy of the SLM imprinting. The other half of the beam passes through the non-linear disordered medium (NLDM) and is captured by the final output camera (CAM). The speckle pattern is then processed by a one-layer feedforward neural network without activation function. The setup was built by our collaborator Luis Mickeler, who kindly provided the image, at ETH Zurich in the group of Prof. Rachel Grange.

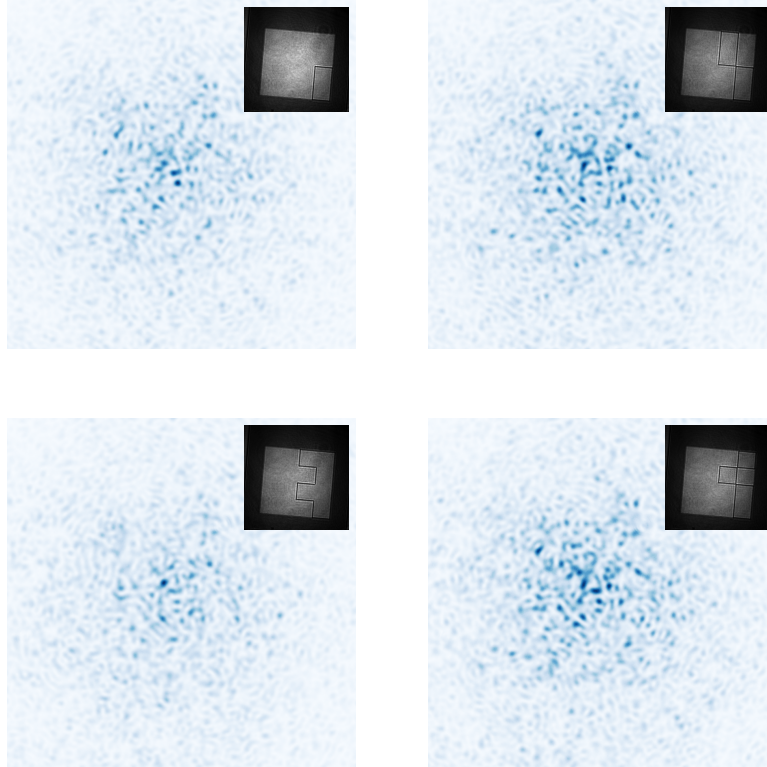


Figure 4.7: Four different basis states $|\sigma_i\rangle$ and their corresponding speckle patterns. The subfigures in the upper right corner show a snapshot of the SLM. All speckle patterns are distinct, because they uniquely correspond to a certain input light field. Courtesy of Luis Mickeler.

5

Results

5.1 GROUND STATE SEARCH

We benchmarked different digital and analogue variational ansätze by comparing their performance in estimating the ground state energies for the transverse field Ising model (TFIM), the Heisenberg model and the J_1 - J_2 Heisenberg model. The parameters of the TFIM and the J_1 - J_2 Heisenberg models were chosen to be at the critical point, which are the most numerically challenging points in the phase diagram. The critical point of the two-dimensional TFIM is at $h/J = 3.04438$ [55] and of the J_1 - J_2 Heisenberg model at $J_1/J_2 = 0.5$ [56]. For more challenging ground states, like the Heisenberg and the J_1 - J_2 Heisenberg model, which have a complex sign structure, the analogue wavefunctions outperform the digital Restricted Boltzmann Machine (RBM) ansatz for the same number of parameters (Fig. 5.1). This demonstrates the higher expressive power of the analogue neural quantum state ansatz when it comes to capturing complex phase relationships and non-trivial correlations.

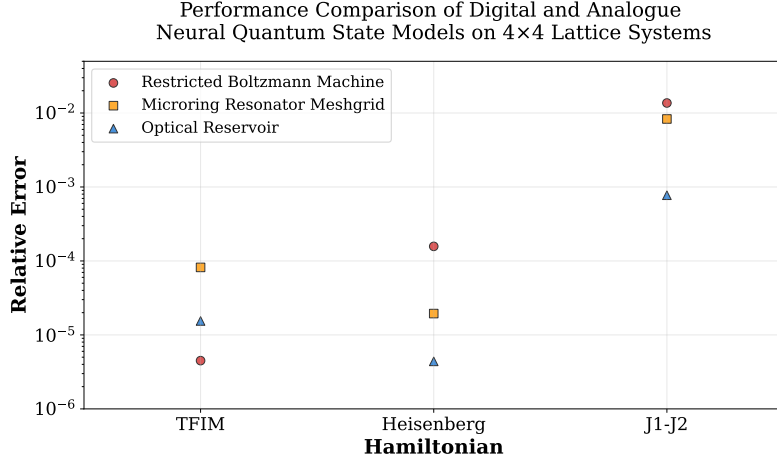


Figure 5.1: Relative error of ground state energies for different Hamiltonians. The J_1 - J_2 Heisenberg model is the most challenging model in terms of sign structure complexity. Ground state energies of digital and analogue wavefunctions are compared to results from exact diagonalization. The RBM obtains the lowest energy for the TFIM, however, for the Heisenberg and the J_1 - J_2 Heisenberg model the analogue wavefunctions reach lower ground states than the RBM by one or two orders of magnitude.

A similar trend is observed for larger system sizes, although the optical reservoir could not yet be tested as further experimental hardware implementations are necessary.

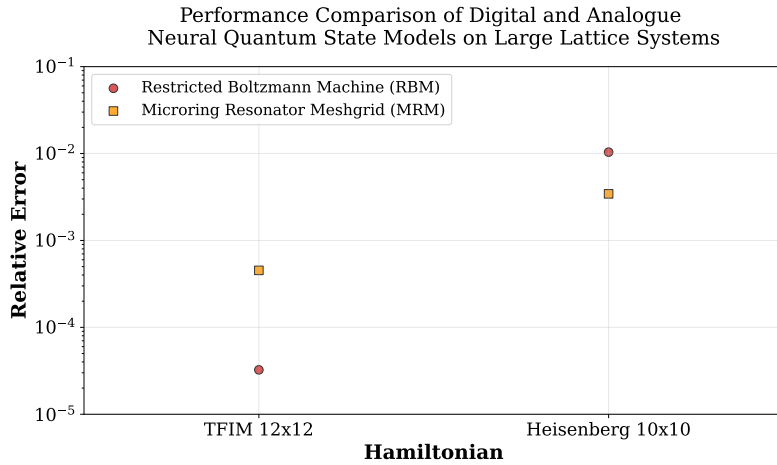


Figure 5.2: Relative error of ground state energies for different Hamiltonians as well as digital and analogue wavefunctions on a 12x12 and 10x10 lattice. Since no exact solution exists for these system sizes, we compared to state of the art results from recurrent neural networks [48]. Again the RBM is more accurate for the TFIM, while the analogue MRM wavefunction reaches a lower ground state energy for the Heisenberg model by one order of magnitude.

Although the RBM is not the current state-of-the-art (SOTA) model for solving these Hamiltonians optimally - see [48] for SOTA results - our findings serve as a proof of concept, showcasing the potential of analogue wavefunctions to exceed standard digital wavefunction parameterizations in accuracy, especially for Hamiltonians with

frustrated sign structures.

Several techniques were employed to improve the stability and accuracy of the ground state search. While some of them will be described in more detail in the following subsections, the Marshall-Peierls sign rule deserves a particular mention. For the Heisenberg model the sign structure of the ground state was known and we used a Marshall-Peierls sign prior [57, 58]. This involves unitarily rotating the Hamiltonian such that the ground state wavefunction amplitudes are non-negative. Effectively, the problem of learning a difficult sign structure - which requires wavefunction expressivity - is removed and the algorithm only has to optimize the amplitudes of the wavefunction, an inherently easier problem.

This technique is, however, less useful near phase transitions of the J_1 - J_2 model, since the ground state is highly frustrated at the critical point. In the latter model the system is in an antiferromagnetic striped order at large J_2 and Néel order for small J_2 , while in the intermediate regime both phases strongly compete, creating high frustration.

To represent sophisticated sign structures, the variational ansatz has to either employ complex-valued parameters and outputs to capture the phase of each wavefunction coefficient or decompose the wavefunction into two parts, modeling the real and imaginary parts separately. Although this is viable, it is twice as costly and generally less performant in practice [59], wherefore we opted for the former approach.

5.1.1 SYMMETRIZATION IMPACT

Variational Wavefunctions can reach lower ground state energies when incorporating physical symmetries [60]. The symmetry constraints restrict the possible wavefunction space to a physically relevant submanifold and thus allow for better and faster optimization. On a lattice with periodic boundary conditions, there is typically translational, rotational and reflection symmetry. In order to include these invariances in the wavefunction, it can be expressed as an average over all possible symmetries to force the ansatz into the desired symmetry sector,

$$\Psi_{sym} = \frac{1}{N} \sum_{g \in G} \Psi(\sigma_g) |\sigma_g\rangle, \quad (5.1)$$

with g an element in the set of symmetries G and N , the number of elements in G .

The following table summarizes the quantitative improvements through symmetrization for the different models and wavefunctions. The improvement factor is defined as the ratio between the ground state energy obtained without and with a symmetrized ansatz.

Table 5.1: Symmetrization impact for different models and wavefunctions.

Symmetrization Improvement Factor	RBM	MRM	OR
TFIM	1	53	1093
Heisenberg	5	5	1155
J_1 - J_2	3	5	863

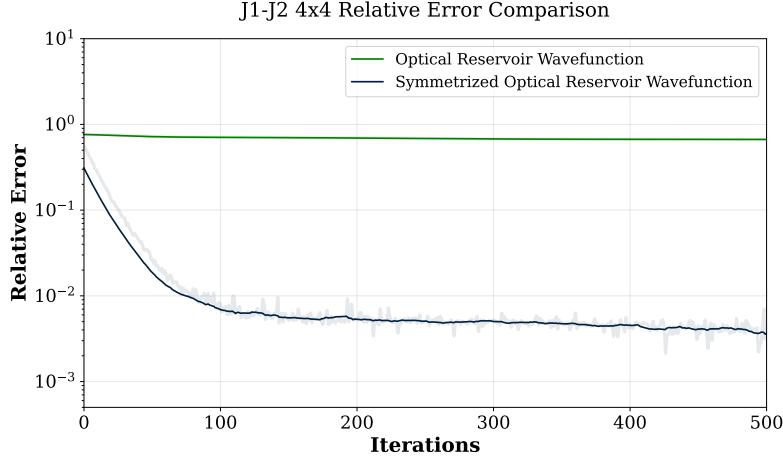


Figure 5.3: Improvement of the ground state energy for the J_1 - J_2 model with 4x4 sites with a symmetric wavefunction ansatz. The optical reservoir wavefunction is an extreme example for the symmetrization improvement factor, because it does not reliably converge to a ground state without symmetrization, due to high experimental noise perturbing the optimization. Instead, ground state search is only possible when averaging over different symmetry sectors and precisely filtering out the noise. The improvement in the relative error of the ground state energy for the MRM and RBM is between a factor of 3 and 53, excluding the RBM for the TFIM, as seen in Table 5.1.

5.1.2 DOWNSAMPLING EFFECTS IN OPTICAL RESERVOIR

The speckle patterns captured by the camera of the optical reservoir consist of roughly 2000 x 2000 pixels. In order to decrease computational cost and remove irrelevant features, the images are cropped to 500 x 500 pixels. Further, the images are downsampled to an even smaller size by averaging neighboring pixels. For optimal downsampling we choose the size of one pixel to correspond to the average size of a speckle, which can be derived from the autocorrelation of the speckle image, as illustrated in Fig. 5.4.

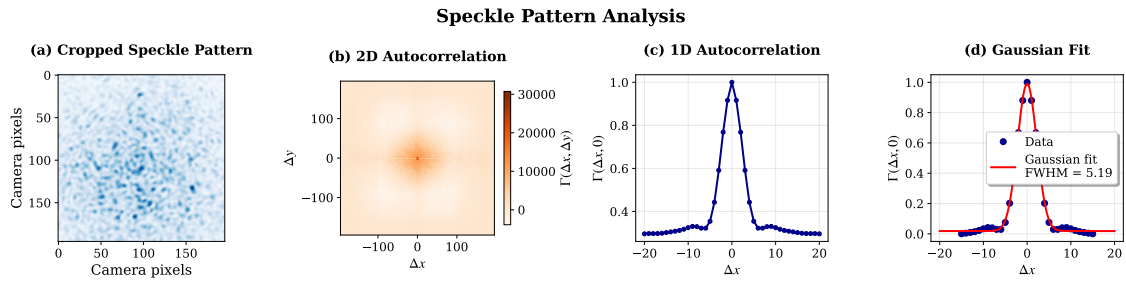


Figure 5.4: Analysis of average speckle size. In a), the speckle pattern is cropped to a certain region of interest. In b), the mean-subtracted cross-correlation $\Gamma(\Delta x, \Delta y)$ of all pixels is computed, where the x and y labels correspond to the distances between the pixels (in units of pixels). In c) a one-dimensional cut $\Gamma(\Delta x, 0)$ of the 2D autocorrelation function along the x-axis is shown. Fitting the autocorrelation function in d) with a Gaussian allows to extract the full width at half maximum (FWHM) and thereby calculate the average size of a speckle. This analysis was performed by our experimental collaborator Luis Mickeler.

Importantly, the dimensionality of the downsampled image conditions the number of trainable parameters in the digital linear layer. It is not surprising that for larger output images and corresponding wider linear layers, the expressivity of the variational model increases and the accuracy of the ground state is improved [50], as shown in Fig. 5.5.

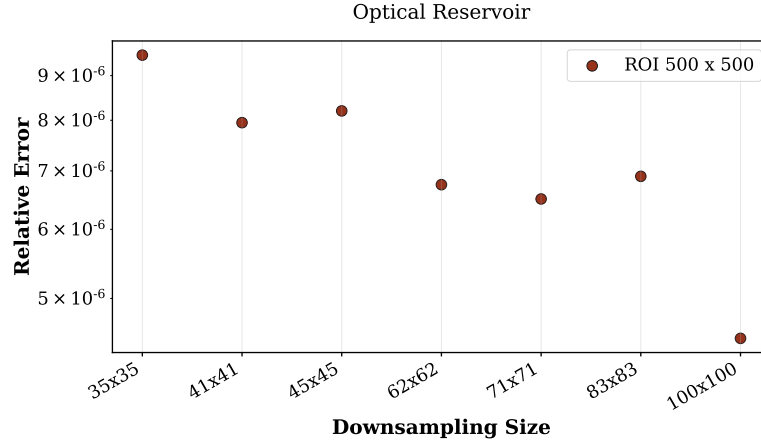


Figure 5.5: Downsampling impact on the relative error of the ground state energy for the TFIM model on a 4x4 lattice.

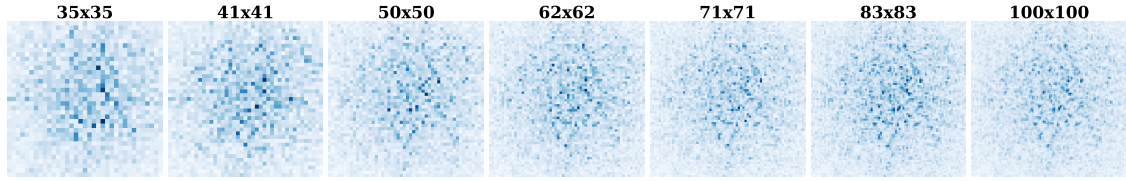


Figure 5.6: Exemplary speckle patterns showing different downsampling sizes. The larger the downsampling size, the more pixels are averaged together, resulting in a coarser image and ultimately a loss of information, leading to a lower accuracy of the ground state energy estimation.

5.1.3 INFLUENCE OF DATA REPLICATION IN MICRORING RESONATOR MESH-GRID

The training stability of the Microring Resonator Meshgrid wavefunction was significantly improved through two data replication techniques. First, the input data was replicated at later layers. As input information propagates through the analogue device, it is transformed according to the internal dynamics and at subsequent layers the same old input is reintroduced and added onto the transformed data, akin to a skip connection in residual blocks [61]. This improves gradient flow and feature reuse and in [29] the authors even argue that this is essential to fit higher order terms (apart from the inverse in the scattering matrix) and thereby enables better generalization.

Second, replication of the input was employed in the initial layer by inserting various copies of the input into

the analogue system. This mechanism effectively introduces a new type of non-linearity, distinct from the inverse in the scattering matrix and increases the input dimensionality to enforce higher order feature generation in the subsequent layers. When input terms x_i interact with copies of themselves through the scattering process, they also create higher order combinations like x_i^2 to x_i^R , depending on the number of copies R . The following figures illustrate the training stabilization and ground state energy improvement.

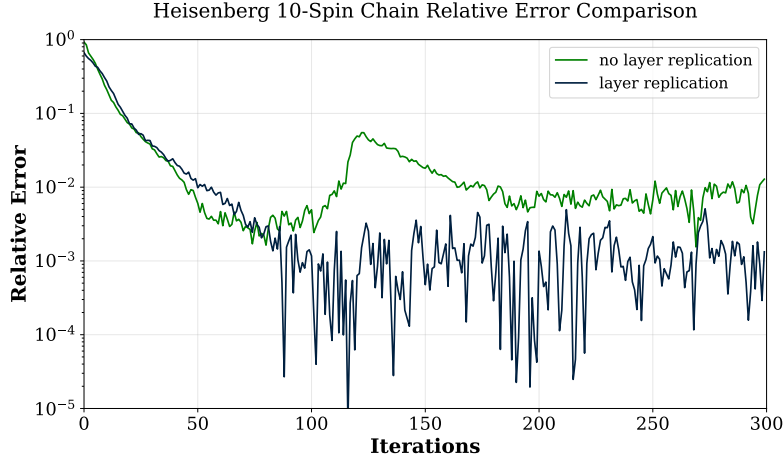


Figure 5.7: Comparison of the optimization behaviour for the Heisenberg model when data replication is used. Two representative training runs are shown. Without data replication the optimization does not reach the ground state, while with data replication a lower ground state energy is reached. The high variance in the blue curve after reaching the ground state is a consequence of the unavoidable sampling noise.

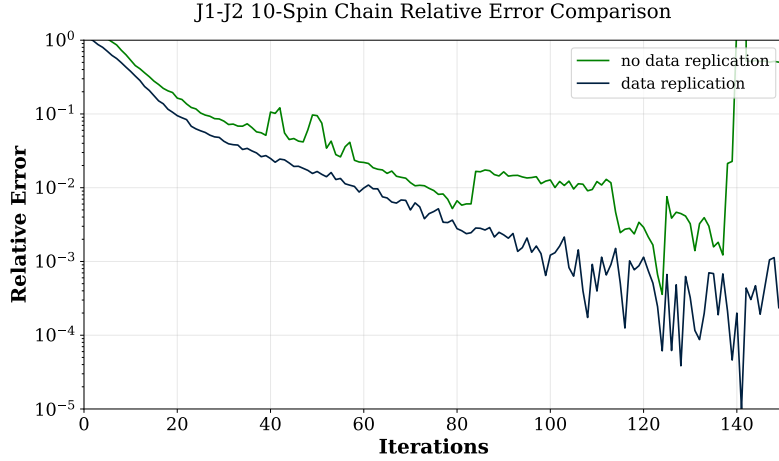


Figure 5.8: Comparison of the optimization behaviour for the J_1 - J_2 model when data replication is used. Two representative training runs are shown. A smoother training process and a lower ground state energy can be observed with data replication. While some runs without data replication reach the same ground state, there are many unwanted and avoidable abrupt jumps to higher energy values in the optimization trajectory, which we can simply avoid via our replication techniques.

5.2 TIME EVOLUTION

In section 3.3 equations were derived governing the time-evolution of quantum states in the framework of Variational Monte Carlo (Eq. 3.19). Here, we apply these equations to simulate a quench of the transverse field of TFIM from $h = 1 \rightarrow 0.5$, which drives the system out of equilibrium. The transverse magnetization $\langle \sigma_x(t) \rangle$ is computed as a function of time. The results are again benchmarked against exact diagonalization calculations for a one-dimensional chain with 8 sites and show excellent agreement.

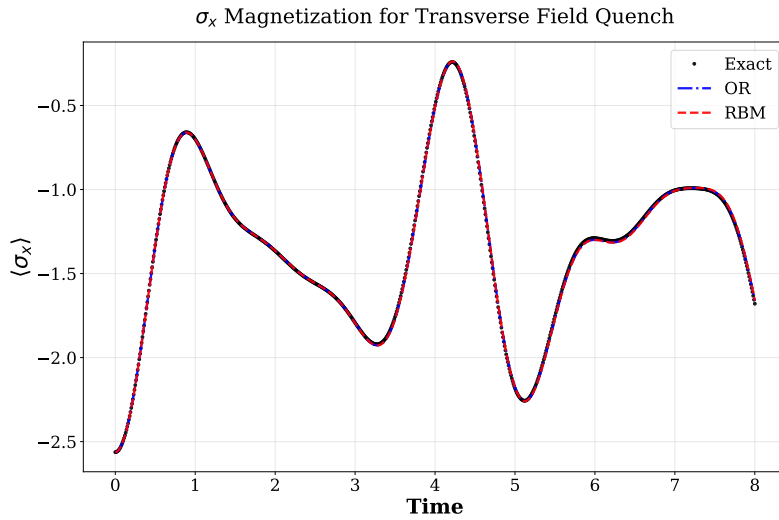


Figure 5.9: Time evolution of the transverse magnetization for the TFIM model with different wavefunction ansätze. The OR and RBM wavefunction dynamics are almost identical to the exact solution. This result was obtained by our theoretical collaborator Vinicius Mohr.

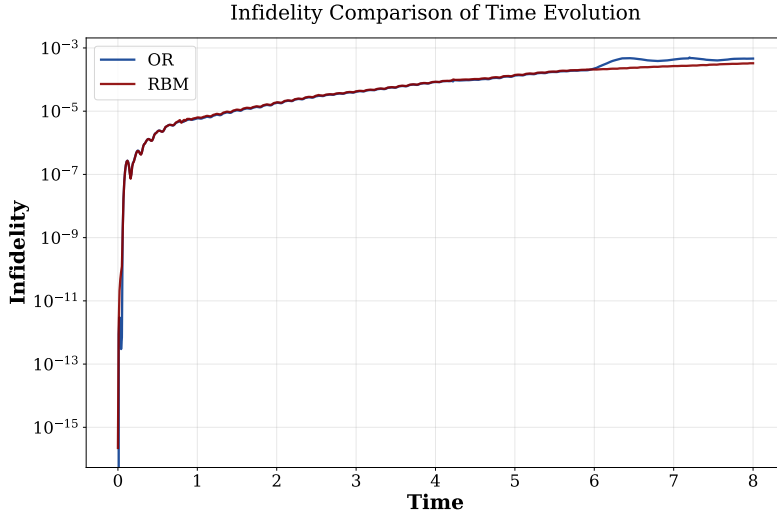


Figure 5.10: Infidelity comparison of time evolution for different wavefunctions compared to the exact solution. The ground states have a similar increase in infidelity with respect to the exact solution. Courtesy of Vinícius Mohr.

5.3 H₂ MOLECULE

Further notable work was done by our theoretical collaborator Vinícius Mohr on solving the intermolecular distance of H₂. By mapping the continuous-space molecule onto a spin system and choosing a truncated orbital basis set (6-31G basis), one can compute ground states for each discretized distance configuration and find the equilibrium bond distance [62].

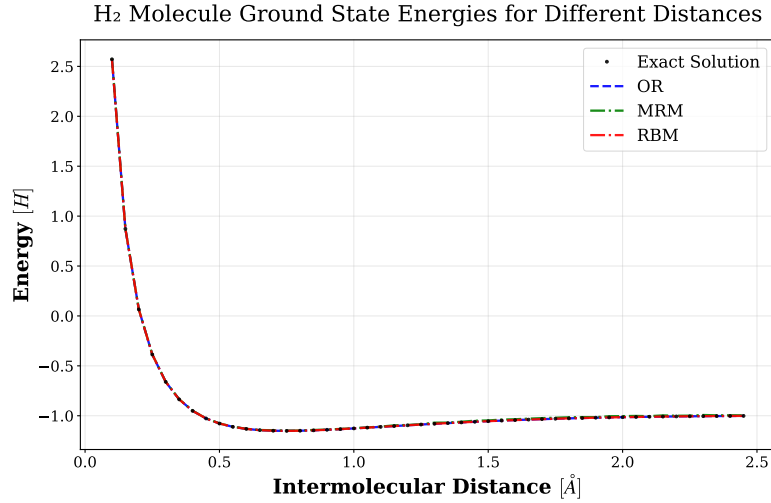


Figure 5.11: Energies for different intermolecular distances for the H₂ molecule with different wavefunction ansätze. The equilibrium bond distance is found at 0.74 Å.

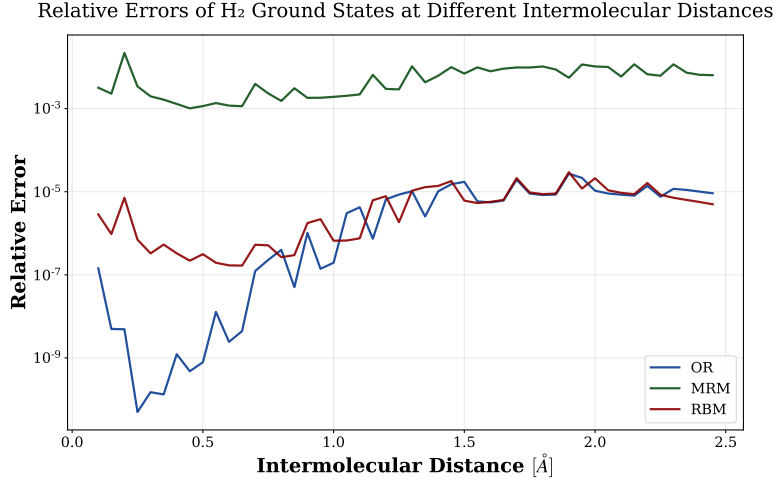


Figure 5.12: Relative errors for different wavefunctions for varying intermolecular distances of the H_2 molecule. The optical reservoir wavefunction reaches lower or equal ground state energies than the RBM wavefunction.

5.4 NON-STABILIZERNESS, ENTANGLEMENT AND FERMIONIC ANTIPLATNESS

Stabilizer entropy, entanglement entropy and fermionic antiplatness [63, 64, 65] are fundamental characteristics of quantum states that are difficult to reproduce classically. Measuring these quantities helps not only to understand the quantum nature of a state, but also the *quantumness* of the analogue devices and thereby implicitly the expressive power of their wavefunction representation. An analysis of these three *quantumness* measures of the different wavefunction ansätze at initialization was performed. The quantumness was analyzed for different initializations at varying standard deviations of the normal distribution from which the initial parameters were sampled from. Non-stabilizeress here refers to the Stabilizer 1-Renyi entropy [64, 65] of the wavefunction.

$$M_\alpha = \frac{1}{1-\alpha} \log \left(\frac{1}{2^N} \sum_{\hat{P} \in \hat{P}_N} \left(\frac{\langle \Psi | \hat{P} | \Psi \rangle}{\langle \Psi | \Psi \rangle} \right)^{2\alpha} \right) \quad (5.2)$$

with $\alpha = 1$ and $\hat{P}_N$, the set of all Pauli operator strings of length N .

Entanglement entropy is the von Neumann entropy of the reduced density matrix of a subsystem.

$$S_{\text{ent}} = -\text{Tr}_A(\rho_A \log \rho_A), \quad (5.3)$$

where $\rho_A = \text{Tr}_{-A}(|\Psi\rangle\langle\Psi|)$ is the reduced density matrix of the subsystem A . Fermionic magic is a measure of the fermionic nature of a state.

$$F = N - \frac{1}{2} \text{Tr}[(M^T M)^k], \quad (5.4)$$

where M is the covariance matrix with elements $M_{i,j} = -\frac{i}{2} \langle\Psi| [\gamma_i, \gamma_j] |\Psi\rangle$ and the Majorana operators γ_i [63].

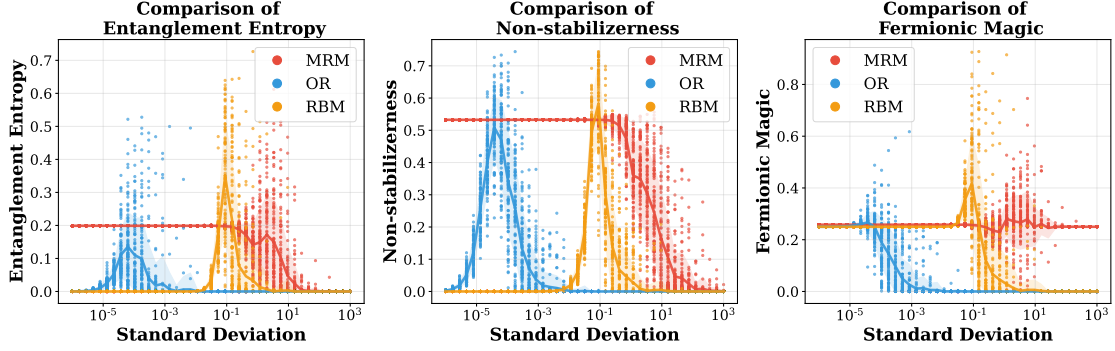


Figure 5.13: Analysis of normalized non-stabilizerness, entanglement entropy and fermionic magic for different wavefunction ansatz initializations. The normalization is achieved by simply dividing by N - or $\frac{N}{2}$ for the entanglement entropy. The point clouds are the results for 100 samples, while the curves are the average values. An extra band around the averages helps to visualize the spread of the data. This study was performed by our theoretical collaborator Vinícius Mohr.

In Fig. 5.13 significant peaks and plateaus can be seen for certain values of the standard deviation. It is not conclusive whether these phase transitions also appear in the thermodynamic limit and whether initializing parameters at points with high magic increases VMC performance for difficultly representable states. For more details we suggest reading [66].

6

Conclusion and Outlook

In this work we demonstrated that analogue computing can be successfully employed as a tool to parametrize quantum many-body wavefunctions. Within the framework of Variational Monte Carlo (VMC) it is possible to compute ground states and dynamics of Hamiltonians across varying levels of complexity. Higher accuracy ground-state energies can be obtained than those produced by the digital Restricted Boltzmann Machine (RBM) ansatz for both the Heisenberg and Heisenberg J_1 - J_2 models. Moreover, the dynamics of small quenched systems can be simulated with equal precision to the RBM ansatz, and other problems, such as determining the equilibrium intermolecular distance of H_2 can be solved with equal or superior accuracy than the RBM ansatz.

In addition, several techniques for the optimization of analogue neural quantum states were presented, including wavefunction symmetrization, downsampling of the optical reservoir speckles and data replication in the microring resonator meshgrid, which were crucial to improve VMC convergence in terms of accuracy and stability.

To the best of our knowledge, this marks the first step towards the integration of a classical analogue wavefunction ansatz in computational physics and, specifically, into the VMC framework. Although many questions have not been addressed yet, like computational speed and scalability, the results establish a promising foundation for the application of analogue devices in addressing complex problems in quantum many-body physics.

Several avenues of future research have to be explored. The first is to extend the current experimental approach of the optical reservoir to simulate larger system sizes. For that purpose, an integrated computational-experimental pipeline should be developed for leveraging the high degree of data throughput, permitting the exploration of previously untapped lattice dimensions in VMC. Further, a systematic analysis of computational speed and en-

ergy consumption is vital in order to evaluate the potential of these specific analogue devices for VMC. While the optical reservoir is a promising candidate in terms of expressivity of the wavefunction and ground state energy performance, its energy consumption remains a major limitation. The second harmonic generation (SHG) process requires a high-power laser beam of 800 mW to generate a sufficiently strong signal in the output. Most of the energy is absorbed by the nonlinear crystal which increases the system’s overall energy demands severely. However, one could investigate replacing the SHG nonlinearity with just the camera’s non-linearity and measure the output at the same wavelength directly to reduce energy consumption. Nevertheless, a careful performance analysis has to be conducted to assess the trade-off between energy consumption and accuracy.

Moreover, the sampling speed could be further improved by implementing an importance sampling scheme, which reduces the number of experimental inference runs needed and therefore overall runtime.

Additional applications of analogue neural quantum states are also currently being investigated. For instance, quantum state tomography using analogue neural quantum states may be a valuable benchmark to illustrate the expressivity and generalization capabilities of the analogue architectures [18]. Furthermore, adding trainable parameters to the optical reservoir, similarly encoded in the phase or amplitude of the incoming light, could remove the need for a final digital layer and yield a fully analogue reservoir wavefunction. However, determining gradients in the fully optical system might present a serious challenge, as chaotic systems like the optical reservoir cannot be modeled well. Nevertheless, there are numerical optimization techniques that are promising, such as direct feedback alignment which has already been successfully applied to optical neural networks [67, 68] or derivative-free optimization, such as finite-difference based optimization or policy gradient methods [69].

In order to increase the expressive power of the optical reservoir wavefunction, one could also explore generating higher order features by cascading multiple non-linear crystals. For example, passing the speckle pattern through additional crystals, analogously to stacking multiple layers in a deep neural network. The increased complexity of the representation may enhance variational energy performance. Beyond higher order features, simply increasing the number of features or different features might also improve performance: in the case of the optical reservoir, less downsampling of the speckle pattern - i.e. using more pixels and information - was shown to increase the accuracy of the ground state, as seen in section 5.1.2. Additionally, the measurement scheme can be extended beyond intensity detection to include phase information of the speckle field through, for example, homodyne detection. A complex-valued output representation carries more information and could be particularly advantageous for modeling ground states with sign structure all-optically, although it may also benefit the learning of other correlated quantum many-body states in general.

References

- [1] A. Einstein, “Zur Elektrodynamik bewegter Körper. (German) [On the electrodynamics of moving bodies],” vol. 322, no. 10, pp. 891–921, 1905.
- [2] J. S. Bell, “On the einstein podolsky rosen paradox,” *Physica Physique Fizika*, vol. 1, no. 3, p. 195, 1964.
- [3] A. A. Michelson and E. W. Morley, “On the relative motion of the earth and the luminiferous ether,” *American journal of science*, vol. 3, no. 203, pp. 333–345, 1887.
- [4] B. Hensen, H. Bernien, A. E. Dréau, A. Reiserer, N. Kalb, M. S. Blok, J. Ruitenberg, R. F. Vermeulen, R. N. Schouten, C. Abellán *et al.*, “Loophole-free bell inequality violation using electron spins separated by 1.3 kilometres,” *Nature*, vol. 526, no. 7575, pp. 682–686, 2015.
- [5] K. R. Popper and G. Weiss, “The logic of scientific discovery,” *Physics Today*, vol. 12, no. 11, pp. 53–54, 11 1959. [Online]. Available: <https://doi.org/10.1063/1.3060577>
- [6] B. Robič, *The Foundational Crisis of Mathematics*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2020, pp. 9–30. [Online]. Available: https://doi.org/10.1007/978-3-662-62421-0_2
- [7] O. Neurath, *Wissenschaftliche Weltanschauung: Der Wiener Kreis*. Dordrecht: Springer Netherlands, 1973, pp. 299–318. [Online]. Available: https://doi.org/10.1007/978-94-010-2525-6_9
- [8] K. Ciesielski, *Set Theory for the Working Mathematician*, ser. London Mathematical Society Student Texts. Cambridge University Press, 1997. [Online]. Available: <https://books.google.ch/books?id=tTEaMFvzhDAC>
- [9] S. Hunklinger, *Festkörperphysik*. Berlin, Boston: De Gruyter Oldenbourg, 2018. [Online]. Available: <https://doi.org/10.1515/9783110567755>
- [10] D. R. Hartree, “The wave mechanics of an atom with a non-coulomb central field. part ii. some results and discussion,” *Mathematical Proceedings of the Cambridge Philosophical Society*, vol. 24, no. 1, p. 111–132, 1928.
- [11] P. Hohenberg and W. Kohn, “Inhomogeneous electron gas,” *Phys. Rev.*, vol. 136, pp. B864–B871, Nov 1964. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRev.136.B864>
- [12] S. R. White, “Density matrix formulation for quantum renormalization groups,” *Phys. Rev. Lett.*, vol. 69, pp. 2863–2866, Nov 1992. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.69.2863>

- [13] U. Schollwöck, “The density-matrix renormalization group in the age of matrix product states,” *Annals of Physics*, vol. 326, no. 1, pp. 96–192, 2011, january 2011 Special Issue. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0003491610001752>
- [14] H. Lange, A. Van de Walle, A. Abedinnia, and A. Bohrdt, “From architectures to applications: a review of neural quantum states,” *Quantum Science and Technology*, vol. 9, no. 4, p. 040501, sep 2024. [Online]. Available: <https://dx.doi.org/10.1088/2058-9565/ad7168>
- [15] G. Carleo and M. Troyer, “Solving the quantum many-body problem with artificial neural networks,” *Science*, vol. 355, no. 6325, pp. 602–606, 2017. [Online]. Available: <https://www.science.org/doi/abs/10.1126/science.aag2302>
- [16] Y. Nomura, N. Yoshioka, and F. Nori, “Purifying deep boltzmann machines for thermal quantum states,” *Phys. Rev. Lett.*, vol. 127, p. 060601, Aug 2021. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.127.060601>
- [17] K. Choo, G. Carleo, N. Regnault, and T. Neupert, “Symmetries and many-body excitations with neural-network quantum states,” *Phys. Rev. Lett.*, vol. 121, p. 167204, Oct 2018. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.121.167204>
- [18] G. Torlai, G. Mazzola, J. Carrasquilla, M. Troyer, R. Melko, and G. Carleo, “Neural-network quantum state tomography,” *Nature Physics*, 2018.
- [19] P. Villalobos, J. Sevilla, T. Besiroglu, L. Heim, A. Ho, and M. Hobbhahn, “Machine learning model sizes and the parameter gap,” *arXiv preprint arXiv:2207.02852*, 2022.
- [20] N. H. Farhat, D. Psaltis, A. Prata, and E. Paek, “Optical implementation of the hopfield model,” *Appl. Opt.*, vol. 24, no. 10, pp. 1469–1475, May 1985. [Online]. Available: <https://opg.optica.org/ao/abstract.cfm?URI=ao-24-10-1469>
- [21] Z. Xu, T. Zhou, M. Ma, C. Deng, Q. Dai, and L. Fang, “Large-scale photonic chiplet taichi empowers 160-tops/w artificial general intelligence,” *Science*, vol. 384, no. 6692, pp. 202–209, 2024. [Online]. Available: <https://www.science.org/doi/abs/10.1126/science.adl1203>
- [22] P. L. McMahon, “The physics of optical computing,” *Nature Reviews Physics*, vol. 5, no. 12, p. 717–734, Oct. 2023. [Online]. Available: <http://dx.doi.org/10.1038/s42254-023-00645-5>
- [23] T. Wang, S. Ma, L. Wright *et al.*, “An optical neural network using less than 1 photon per multiplication,” *Nature Communications*, vol. 13, no. 1, p. 123, 2022.
- [24] A. Lugt, “Signal detection by complex spatial filtering,” *IEEE Transactions on Information Theory*, vol. 10, no. 2, pp. 139–145, 1964.
- [25] N. H. Farhat, D. Psaltis, A. Prata, and E. Paek, “Optical implementation of the hopfield model,” *Appl. Opt.*, vol. 24, no. 10, pp. 1469–1475, May 1985. [Online]. Available: <https://opg.optica.org/ao/abstract.cfm?URI=ao-24-10-1469>
- [26] H.-Y. S. Li, Y. Qiao, and D. Psaltis, “Optical network for real-time face recognition,” *Appl. Opt.*, vol. 32, no. 26, pp. 5026–5035, Sep 1993. [Online]. Available: <https://opg.optica.org/ao/abstract.cfm?URI=ao-32-26-5026>

- [27] D. A. B. Miller, “Self-configuring universal linear optical component
invited
,” *Photon. Res.*, vol. 1, no. 1, pp. 1–15, Jun 2013. [Online]. Available: <https://opg.optica.org/prj/abstract.cfm?URI=prj-1-1-1>
- [28] J. Spall, X. Guo, T. D. Barrett, and A. I. Lvovsky, “Fully reconfigurable coherent optical vector-matrix multiplication,” *Optics Letters*, vol. 45, no. 20, p. 5752, Oct. 2020. [Online]. Available: <http://dx.doi.org/10.1364/OL.401675>
- [29] C. C. Wanjura and F. Marquardt, “Fully nonlinear neuromorphic computing with linear wave scattering,” *Nature Physics*, vol. 20, no. 9, pp. 1434–1440, 2024.
- [30] J. Dong, M. Rafayelyan, F. Krzakala, and S. Gigan, “Optical reservoir computing using multiple light scattering for chaotic systems prediction,” *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 26, no. 1, pp. 1–12, 2020.
- [31] G. Wetzstein, A. Ozcan, S. Gigan, S. Fan, D. Englund, M. Soljačić, C. Denz, D. Miller, and D. Psaltis, “Inference in artificial intelligence with deep optics and photonics,” *Nature*, vol. 588, pp. 39–47, 12 2020.
- [32] T. Ando, R. Horisaki, and J. Tanida, “Speckle-learning-based object recognition through scattering media,” *Opt. Express*, vol. 23, no. 26, pp. 33 902–33 910, Dec 2015. [Online]. Available: <https://opg.optica.org/oe/abstract.cfm?URI=oe-23-26-33902>
- [33] H. Wang, J. Hu, A. Morandi, A. Nardi, F. Xia, X. Li, R. Savo, Q. Liu, R. Grange, and S. Gigan, “Large-scale photonic computing with nonlinear disordered media,” 2023. [Online]. Available: <https://arxiv.org/abs/2310.07690>
- [34] L. Wright, “Invited talk ”deep physical neural networks” at the conference on computing with physical systems in january 2024,” 2024. [Online]. Available: <https://www.youtube.com/watch?v=ryFAB4Ko6Vw>
- [35] M. A. Nahmias, T. F. de Lima, A. N. Tait, H.-T. Peng, B. J. Shastri, and P. R. Prucnal, “Photonic multiply-accumulate operations for neural networks,” *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 26, no. 1, pp. 1–18, 2020.
- [36] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed. Cambridge University Press, 2018.
- [37] E. Schrödinger, “Quantisierung als eigenwertproblem, erste mitteilung,” *Ann. Phys.*, vol. 79, pp. 361–376, 1926.
- [38] J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*, 3rd ed. Cambridge University Press, 2020.
- [39] A. Weiß and H. Fehske, *Exact Diagonalization Techniques*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 529–544. [Online]. Available: https://doi.org/10.1007/978-3-540-74686-7_18
- [40] D. P. A. M., “Quantum mechanics of many-electron systems,” *Proc. R. Soc. Lond*, 1929.

- [41] D. Ceperley, G. V. Chester, and M. H. Kalos, “Monte carlo simulation of a many-fermion study,” *Phys. Rev. B*, vol. 16, pp. 3081–3099, Oct 1977. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevB.16.3081>
- [42] J. A. Barker, “A quantum-statistical monte carlo method; path integrals with boundary conditions,” *The Journal of Chemical Physics*, vol. 70, no. 6, pp. 2914–2918, 03 1979. [Online]. Available: <https://doi.org/10.1063/1.437829>
- [43] G. Carleo, *Computational Quantum Physics Lecture Notes*. École Polytechnique Fédérale de Lausanne, 2024.
- [44] S. Sorella, “Green function monte carlo with stochastic reconfiguration,” *Phys. Rev. Lett.*, vol. 80, pp. 4558–4561, May 1998. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.80.4558>
- [45] E. Ising, “Beitrag zur theorie des ferromagnetismus,” *Z. Physik*, vol. 31, pp. 253–258, feb 1925. [Online]. Available: <https://doi.org/10.1007/BF02980577>
- [46] F. Becca and S. Sorella, *Quantum Monte Carlo Approaches for Correlated Systems*. Cambridge University Press, 2017.
- [47] J. Hubbard, “Electron correlations in narrow energy bands,” *Proc. R. Soc. Lond. A*, vol. 276, pp. 238–257, 1963.
- [48] D. Wu, R. Rossi, F. Vicentini, N. Astrakhantsev, F. Becca, X. Cao, J. Carrasquilla, F. Ferrari, A. Georges, M. Hibat-Allah, M. Imada, A. M. Läuchli, G. Mazzola, A. Mezzacapo, A. Millis, J. R. Moreno, T. Neupert, Y. Nomura, J. Nys, O. Parcollet, R. Pohle, I. Romero, M. Schmid, J. M. Silvester, S. Sorella, L. F. Tocchio, L. Wang, S. R. White, A. Wietek, Q. Yang, Y. Yang, S. Zhang, and G. Carleo, “Variational benchmarks for quantum many-body problems,” *Science*, vol. 386, no. 6719, pp. 296–301, 2024. [Online]. Available: <https://www.science.org/doi/abs/10.1126/science.adg9774>
- [49] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbdo53c1c4a845aa-Paper.pdf
- [50] K. Hornik, M. Stinchcombe, and H. White, “Multilayer feedforward networks are universal approximators,” *Neural Networks*, vol. 2, no. 5, pp. 359–366, 1989. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0893608089900208>
- [51] G. Carleo, I. Cirac, K. Cranmer, L. Daudet, M. Schuld, N. Tishby, L. Vogt-Maranto, and L. Zdeborová, “Machine learning and the physical sciences,” *Rev. Mod. Phys.*, vol. 91, p. 045002, Dec 2019. [Online]. Available: <https://link.aps.org/doi/10.1103/RevModPhys.91.045002>
- [52] Y. Kwak, W. J. Yun, J. P. Kim, H. Cho, M. Choi, S. Jung, and J. Kim, “Quantum distributed deep learning architectures: Models, discussions, and applications,” 2022. [Online]. Available: <https://arxiv.org/abs/2202.11200>

- [53] A. Nardi, A. Morandi, R. Pierrat, A. Goetschy, X. Li, F. Scheffold, and R. Grange, “Mesoscopic light transport in nonlinear disordered media,” *Phys. Rev. Res.*, vol. 7, p. 023323, Jun 2025. [Online]. Available: <https://link.aps.org/doi/10.1103/ztvj-rc63>
- [54] P. Caramazza, O. Moran, R. Murray-Smith, and D. Faccio, “Transmission of natural scene images through a multimode fibre,” *Nature Communications*, vol. 10, no. 1, May 2019. [Online]. Available: <http://dx.doi.org/10.1038/s41467-019-10057-8>
- [55] H. W. J. Blöte and Y. Deng, “Cluster monte carlo simulation of the transverse ising model,” *Phys. Rev. E*, vol. 66, p. 066110, Dec 2002. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevE.66.066110>
- [56] L. Capriotti and S. Sorella, “Spontaneous plaquette dimerization in the $j_1 - j_2$ heisenberg model,” *Phys. Rev. Lett.*, vol. 84, pp. 3173–3176, Apr 2000. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.84.3173>
- [57] L. CAPRIOTTI, “Quantum effects and broken symmetries in frustrated antiferromagnets,” *International Journal of Modern Physics B*, vol. 15, no. 12, pp. 1799–1842, 2001. [Online]. Available: <https://doi.org/10.1142/S0217979201004605>
- [58] F. Becca, L. Capriotti, A. Parola, and S. Sorella, *Variational Wave Functions for Frustrated Magnetic Models*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2011, pp. 379–406. [Online]. Available: https://doi.org/10.1007/978-3-642-10589-0_15
- [59] L. L. Viteritti, F. Ferrari, and F. Becca, “Accuracy of restricted Boltzmann machines for the one-dimensional $J_1 - J_2$ Heisenberg model,” *SciPost Phys.*, vol. 12, p. 166, 2022. [Online]. Available: <https://scipost.org/10.21468/SciPostPhys.12.5.166>
- [60] K. Choo, G. Carleo, N. Regnault, and T. Neupert, “Symmetries and many-body excitations with neural-network quantum states,” *Phys. Rev. Lett.*, vol. 121, p. 167204, Oct 2018. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.121.167204>
- [61] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” 2015. [Online]. Available: <https://arxiv.org/abs/1512.03385>
- [62] M. Qing and W. Xie, “Use vqe to calculate the ground energy of hydrogen molecules on ibm quantum,” 2023. [Online]. Available: <https://arxiv.org/abs/2305.06538>
- [63] P. Sierant, P. Stornati, and X. Turkeshi, “Fermionic magic resources of quantum many-body systems,” 2025. [Online]. Available: <https://arxiv.org/abs/2506.00116>
- [64] L. Leone, S. F. E. Oliviero, and A. Hama, “Stabilizer rényi entropy,” *Phys. Rev. Lett.*, vol. 128, p. 050402, Feb 2022. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevLett.128.050402>
- [65] A. Sinibaldi, A. F. Mello, M. Collura, and G. Carleo, “Non-stabilizerness of neural quantum states,” 2025. [Online]. Available: <https://arxiv.org/abs/2502.09725>
- [66] A. Jreissaty, H. Zhang, J. C. Quijano, J. Carrasquilla, and R. Wiersema, “Entanglement and optimization within autoregressive neural quantum states,” *arXiv preprint arXiv:2509.12365*, 2025.

- [67] M. J. Filipovich, Z. Guo, M. Al-Qadasi, B. A. Marquez, H. D. Morison, V. J. Sorger, P. R. Prucnal, S. Shekhar, and B. J. Shastri, “Silicon photonic architecture for training deep neural networks with direct feedback alignment,” *Optica*, vol. 9, no. 12, pp. 1323–1332, Dec 2022. [Online]. Available: <https://opg.optica.org/optica/abstract.cfm?URI=optica-9-12-1323>
- [68] Z. Wang, K. Müller, M. Filipovich, J. Launay, R. Ohana, G. Pariente, S. Mokaadi, C. Brossollet, F. Moreau, A. Cappelli, I. Poli, I. Carron, L. Daudet, F. Krzakala, and S. Gigan, “Streamlined optical training of large-scale modern deep learning architectures with direct feedback alignment,” 2025. [Online]. Available: <https://arxiv.org/abs/2409.12965>
- [69] J. Yao, M. Bukov, and L. Lin, “Policy gradient based quantum approximate optimization algorithm,” *ArXiv*, vol. abs/2002.01068, 2020. [Online]. Available: <https://api.semanticscholar.org/CorpusID:211020928>

Acknowledgments

First and foremost, I would like to thank my external and internal advisors Prof. Juan Carrasquilla-Alvarez and Prof. Lode Pollet for giving me the opportunity to work on this exciting interdisciplinary topic with them. I am grateful for the freedom they gave me in exploring new ideas and their invaluable feedback and support. Through this journey I have found a true inspiration to pursue research as a career.

Further, I want to express my gratitude towards my close collaborator and great friend Vinícius Mohr, who I have pondered virtually all ideas of this project with. Likewise, I have to thank our experimental colleagues Luis Mickeler, Dr. Alfonso Nardi and Prof. Rachel Grange, without whom this work would not have been possible. And of course, all members of the group of Prof. Carrasquilla-Alvarez, including the cooperative Dr. Jannes Nys, who were part of this journey and contributed to the amazing, open group culture.

Declaration

I hereby declare that this thesis is my own work, and that I have not used any sources and aids other than those stated in the thesis.

Zurich, 31st October 2025

Author's signature